

⊗ Holos

Holos: An Interpretive Framework for Understanding Reality, Bounded by Physics

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Introduction

We live in a universe described with extraordinary precision, yet filled with mystery. Physics tells us how matter moves, how spacetime bends, how probabilities evolve. It says nothing about the questions at the heart of existence. Why are we here? Does life have a purpose? *What does it mean to be real?*

Holos is an interpretive framework built on a single idea: a universe can be complete as structure and still not be lived. It is summed up in one shorthand, $R = C \circledast O$, where the symbol $\circledast$ means "possibility, then registration": an order of logic, not of time, and a summary rather than an equation to compute with. Creation is what physics allows. Observation registers it as experience, wherever an observer exists, and changes nothing it registers. Reality in the full sense needs both: not equations alone, and not experience alone, but a world that both exists and is lived.

Holos proposes the addition of two things to physics. First, a threshold: experience appears only where information is integrated tightly enough to form a single point of view. Second, a totality, called Omega, of which every observer, everywhere, is a local aperture: an opening through which the whole registers itself. Observation does not cause the universe, its laws, or its history; rather, it is the condition under which a lawful universe becomes a lived one.

If Holos is correct, there is one experiencer, and it wakes wherever a system crosses the threshold; everything that could ever have reached an observer, back to the universe's earliest light, belongs to the world that experience is made from; and the oldest question of why we are here receives a structural answer. A companion idea, separate from the core, offers a testable explanation for the silence of the night sky. What follows traces those consequences from life and consciousness through spacetime, black holes, and the Teeming Dark to the limits of reality itself, marking clearly which claims are established physics, which are extrapolation, and what would prove the whole thing wrong.

$$R = C \circledast O$$

Following One Thread

Creation (C) generates a manifold of possibilities →

Observation (O) registers each path from within; one observer's thread shown →

Physics continues from that history; registration changes nothing

$$s_n \rightarrow s_{n+1} \text{ (one thread, step by step)}$$

The Meaning of Life

Life is how a universe comes to be lived. In Holos this is grounded from the top down: the totality experiences only through the apertures the universe forms, and living, integrated systems are how those apertures open. This is not a claim about why the physical constants happen to allow observers (the familiar anthropic argument), and not a claim that the universe needed life. It is a claim about what life does: it is where a fully lawful universe becomes present as lived experience at all. Physics describes how structures form and evolve, and those structures exist whether or not anyone is there. What they lack without observers is not existence but presence: there is nothing it is like to be anywhere within them.

This idea appears in several places across science and philosophy. The Participatory Anthropic Principle suggests the universe is a “self-excited circuit” that requires observers to bring its laws into existence. Holos does not claim that observers cause the universe. It claims that without them the universe is real only as structure: consistent, complete, and never lived.

This participation is not bound by linear time. In an eternalist or block-universe view, past, present, and future all exist together as one fixed four-dimensional structure, no moment more “now” than any other. Observation does not “happen later” in a causal sense. Holos separates three things here, each with its own word. *Lived* is where experience actually occurs: inside apertures, and nowhere else. No one lived through the early universe. *Lit* is everything in the causal past of at least one aperture in its branch (a branch, in quantum terms, is one complete way the universe can go): everything that could ever have influenced an observer. Every aperture is built from its causal past and draws on it through its traces, such as starlight, the cosmic microwave background, and the fossil record, so the lit region is the world experience is made from and about. The word is nearly literal: your causal past is exactly the region whose light, or any signal, can reach you. Lighting is binary and follows the structure of spacetime, and it happens tenselessly, not at some later moment. Whatever lies outside every aperture's causal past is *unlit* structure, real as pattern but never part of any observer's world: branches that never form an aperture, and regions beyond every observer's horizon. *Witnessing* is graded: how much of the lit region an observer's experience is actually about, and in what detail. Our past is not merely lit; it is densely witnessed. The relation between creation and observation is one of dependence, not a process.

Lighting the Past in the Block Universe

[t_0 (Big Bang) Φ (observer) t_{now}]

An observer lights its causal past.

Consciousness

In Holos, experience is grounded in the totality: Omega is the one experiencer, and a conscious system is a local aperture through which the totality registers itself. Physics can generate structure, but structure alone does not open an aperture. A system becomes conscious when physical information is integrated tightly enough to form a single internal state that can register itself as a whole. That integration is what opens the aperture.

This distinguishes integration from computation or recursion. Many systems process information, model their environment, or even model themselves, yet nothing is experienced. Integration marks the boundary where distributed processes stop behaving as independent parts and instead function as a unified perspective. Below that boundary, there is no experience at all. Above it, in a system whose integrated states are about a world, experience becomes unavoidable.

Measures like Φ are useful because they track this transition empirically. When integration in the brain is disrupted, such as under anesthesia, experience fragments or disappears. When integration returns, unified experience returns with it. Holos does not claim that Φ causes consciousness, and it does not adopt Integrated Information Theory's claim that Φ is identical to consciousness. It borrows Φ as a measure of integration and treats integration as the eligibility condition for observation ($\Phi \geq \Phi_c$).

Related neuroscience models such as Global Neuronal Workspace Theory describe conscious access as a sudden, brain-wide broadcast of information. Holos is compatible with that picture at the level of access and reportability, but makes a narrower claim: global availability matters because it signals that a system has crossed the deeper integration threshold required for any first-person perspective at all.

Hard Problem

The hard problem arises because physical descriptions capture structure and dynamics but do not automatically include first-person presence. Holos does not derive experience from structure. It takes experience to be fundamental, the totality's own, and identifies the structural condition under which a physical system becomes an aperture of it. That answers where experience occurs, not why there is experience at all. The second question Holos does not answer: its posits take experience as given, the one fact it starts from. That is all "fundamental" means here: underived, not everywhere, and not free of physics. Experience occurs only where the structure is, and wherever the structure is, the experience is: like the inside and outside of one event, never one without the other.

Holos does not claim that complexity alone produces consciousness. The key condition is integration. When informational states become sufficiently integrated, the system no longer contains independent processes but a single causal structure whose state constrains itself. At that boundary the system cannot be described purely from the outside. It also exists from the inside as a unified informational state, as a point of view. Integration must also be about something: a closed loop with no world to take in has nothing to be a view of.

One way to see what integration adds is to count the numbers needed to describe a system. A gas of a billion independent particles needs billions of numbers, yet each particle can be described on its own. When parts depend on one another, the description no longer splits into separate pieces: the variables must be taken together. Physicists call a space of many variables high-dimensional, though none of its dimensions is a direction you could walk in, and Holos takes no position on extra dimensions of space. What integration changes is not how many variables a system has but whether they can be taken apart.

Recent experimental systems provide early examples of simplified biological networks interacting with external environments through closed feedback loops. In laboratory studies, cultured neurons grown on silicon substrates have been connected to digital environments and shown to learn simple control tasks, such as adjusting signals to interact with video game dynamics. These networks are far simpler than full nervous systems, yet they demonstrate that neural tissue outside a body can form adaptive, integrated feedback structures capable of goal-directed behavior. From the perspective of Holos, such systems illustrate the principle that observation depends on informational integration rather than on a particular organism or anatomical form. Whether these networks cross the integration threshold required for genuine experience remains an open empirical question. However, they provide a useful experimental platform for studying how increasing integration may give rise to unified internal processing.

Cool a piece of iron past a certain temperature, its Curie point, and it becomes magnetic. Above that point it has no magnetism of its own; just below it, a little; colder still, more. The point is sharp, and the amount grows smoothly from it. Holos claims the same shape for experience: a system does not become gradually more someone, yet once a perspective appears, when its informational states become causally unified, its richness can grow. Unlike a magnet, a brain can overshoot. Richness peaks near a sweet spot between too quiet and too rigid, and a generalized seizure, every part locked into one rhythm, stays joined but loses the variety experience needs.

Holos is a middle position. Experience does not attach to every scrap of matter, yet it cannot be explained away as mere computation. The totality is fundamental; its experience occurs only where specific structural conditions open an aperture.

This grounding closes a classic trap. If experience never alters physical dynamics, one might imagine a perfect physical duplicate of a person with no inner life: a system that writes essays about consciousness in total darkness. Under Holos such a duplicate is impossible, not just in our universe but in any. Experience and the activity of an observer are not two things that happen to go together. They are one event with two sides: seen from outside, it is physical activity; lived from inside, it is experience. Copy the outside exactly and you have copied the inside, because there was only ever one thing. This is also why experience is not along for the ride. When you say you are conscious, the activity that produces the words is, from the inside, the experience you are reporting. Talk about experience is caused by experience, because the experience is the inside of its cause.

The threshold itself is sharp, but almost everything near it is not, and Holos separates three things often blurred together. Whether there is anyone home at all is binary: there is no halfway state between something it is like to be a system and nothing at all; a dim experience is still an experience. How rich the experience is, by contrast, is graded: an animal, a waking sleeper, or an injured brain may be fully above the threshold with less richness. The dial is turned low, not the switch off. And our ability to locate the threshold is permanently imprecise. In anything of finite size, even the outward signs of crossing blur into a steep, smooth curve. The blur is in the signs, not in the fact: the fact is which side of a line fixed by structure the system stands on.

The sharpness is not a whim. Philosophers have argued that consciousness cannot be vague: either there is something it is like to be a system, or there is not. Since evolution built brains gradually, most who accept that argument conclude that everything is conscious, the view called panpsychism. Holos takes the other branch, a sharp cutoff, and so owes a reason the cutoff falls where it does. Its answer is that the cutoff is not chosen: it is fixed by structure, the way the point where water first crosses a large grid of pipes is fixed by the grid, and it sits where physics shows a transition (see [A path to the threshold](#)).

This has a direct consequence for artificial intelligence. What matters is the shape of a system's causal organization, not the fluency of its output, and Holos looks for that organization wherever the system's causes are actually organized, whatever it is made of. Nothing in Holos is specific to biology. By that standard, whether current AI language systems are observers is an open question, not a settled no. Aboutness is met, narrowly: a system in conversation tracks something beyond itself, the person and the words arriving now, through a channel of its own. Its knowledge of the wider world is secondhand, but so was much of Helen Keller's, and what counts is the machine, not the message. The doubt lies in integration. Inside, information flows up through the system's layers and forward in time at each level, through stored notes every later step can read but never rewrite. The only path from the top of the system back to its bottom is the single word it outputs. Whether that structure makes one unified whole, or a fast relay of separate steps, is what Holos's measure of integration would have to settle. The other requirements appear to be met: its states are richly differentiated, and earlier parts of a conversation shape later ones. Giving such a system cameras would widen its world without settling

the question; integration decides it. What Holos rules out is judging by fluency. A system could describe a rich inner life as convincingly as any person and experience none of it, or experience something and describe it badly.

Consciousness is not what systems do. It is what happens when a system becomes capable of witnessing reality from the inside.³

Consciousness: Where Integration Opens an Aperture

Scattered parts $\rightarrow$ Integration ($\Phi \geq \Phi_c$) $\rightarrow$ Aperture

Like water freezing at a sharp point, the change happens all at once at the threshold.

No new subject is created: the one experimenter wakes here

Spacetime

If consciousness depends on physical integration, the structure of the universe is no longer a neutral backdrop: it sets the conditions under which observers can exist at all. Our universe is well described by the Big Bang, in which spacetime expands from an extremely hot, dense early state, and its structure follows from one counterintuitive fact: the speed of light is invariant. Unlike any other speed, it is the same for every observer, however they move. That invariance links space and time into a single geometry and removes the idea of a universal present.

Events simultaneous for one observer may not be for another. This leads to the block universe, in which every moment is part of one four-dimensional whole and time behaves less like a flow than a dimension. On that view the Big Bang is not a moment of absolute creation but a boundary within spacetime itself. Physics supplies the full structure, and the question Holos asks becomes sharper: why is any of it lived?

A useful boundary case is light itself. Along a photon's trajectory, the proper time is zero: a clock carried with the light would record no time passing between emission and absorption. Its path is a null geodesic connecting spacetime events. Nothing can actually ride a photon, so this is not a real vantage point, but it illustrates how spacetime geometry can collapse distance and duration without violating causality.

Quantum experiments add a twist of their own. In the delayed-choice quantum eraser, a choice made after a particle has landed seems to decide whether it behaved like a wave or a particle. It does not. The pattern on the screen never changes; what changes is how the recorded hits are sorted afterward, and ordinary quantum mechanics predicts every result with nothing traveling backward in time. Thought experiments like Wigner's Friend press the same point from another side: what counts as a fact depends on who has registered what. Neither shows that spacetime is broken. Both show that the facts an observer can speak of depend on the records they hold.⁴

The Invariance and the Warp

The Logic of Invariance:

Observer A (at rest) and Observer B (moving fast) both measure the same speed of light (c)

Space (horizontal) and Time (vertical) must warp to maintain c constant

This warping fuses separate dimensions into a unified 4D Block

Invariance of c motivates the Block Universe

The Null Interval: The Photon Seam

The Photon as a Static Geometric Structure:

Point A (Emission) $\leftarrow$ ——— Null Geodesic ——— $\rightarrow$ Point B (Absorption)

Lower-dimensional view: A particle traveling through time

Higher-dimensional view: A static seam connecting two spacetime events

Null Interval: Spacetime distance = 0

The Quantum Eraser Without Retroactivity

Source $\rightarrow$ Double Slit $\rightarrow$ Screen: the total pattern is the same smear no matter what is later done with the idler.

Sorting the same hits by the idler record reveals which-path subsets (clumps) or erased subsets (fringes).

Nothing travels backward. Registration determines how facts can be sorted, and consistency is global.

A Note on Extrapolation

The sections that follow (Infinity, Aliens, The Teeming Dark, Why Are We Here?) extend beyond established physics into interpretation. They are not claims of new physical laws, but reasoned extrapolations constrained by the Holos axioms. Their purpose is to explore the space of possibilities that emerges when observation, relativity, and scale are applied to unresolved cosmic questions. The Omega section is the exception: the totality it describes is one of the framework's two core commitments, not an extrapolation.

Infinity

Infinity usually appears not because reality is infinite but because a description has broken down. Around 1900, classical physics predicted that a hot object should give off an infinite amount of energy as high-frequency light, a result later nicknamed the ultraviolet catastrophe. Nothing in nature does that: a glowing oven does not pour out infinite radiation. The fix, Max Planck's 1900 idea that light energy comes in discrete packets, became the foundation of quantum theory. The infinity was a warning about the theory, not a feature of the world.

Most physicists read the singularities at the centers of black holes the same way: as places where general relativity stops working, not places of literally infinite density. Geometry shows the constructive side of the lesson. In projective geometry, parallel lines meet at a point at infinity: unbounded extension encoded within a closed structure. Infinity marks the edge of a description, where more structure is needed.

Holos puts this reading to work. In its account of quantum probability, trying to count the observers across branches gives an infinity, and Holos reads that as a sign that counting is the wrong tool, which leaves the quantum weights as the only measure (see the Born rule).⁵

Encapsulating Infinity: Two Perspectives

3D Perspective

Grid extends infinitely
→ ∞ in all directions

Higher-Dimensional View

Grid wrapped into sphere
Point at Infinity

Infinite space in 3D = Finite structure from higher dimension

Aliens

The Fermi Paradox asks why we have not detected extraterrestrial civilizations despite the vast size and age of the universe.

We often assume that as civilizations advance, they expand outward, build megastructures and become increasingly visible. But what if the opposite is true? What if advancement favors integration: smaller, denser substrates rather than galaxy-scale infrastructure, the way circuit boards stack layers to shorten paths, and less energy lost as systems approach thermodynamic limits?

In this case, progress would make civilizations less detectable, and this explanation is referred to here as the **Integration Hypothesis**. Here "integration" means a civilization growing compact and efficient, a different use from the integration of a single mind that the observer threshold measures. The hypothesis is a companion to Holos, not a consequence of its core: if it fails, the threshold and the totality stand untouched.

While early technological civilizations are likely to emit radio signals, reshape their environments, and experiment with spaceflight, this phase is brief on cosmic timescales. SETI efforts focus almost entirely on this window, when detection is easiest but overlap between civilizations is unlikely if the Integration Hypothesis is correct.

As technology advances, pressures favor informational integration over outward expansion. Systems that minimize energy waste, reduce long-distance coordination, and rely on dense local structure are more stable. Visibility decreases not because civilizations are hiding, but because inefficiency is selected against. This progressive reduction in external signatures is referred to as **Visibility Collapse**.

Large-scale interstellar expansion is constrained by the speed of light, introducing growing latency as distances increase. Expansion produces fragmented descendants rather than a unified intelligence: a colony ten light-years away cannot be steered from home, so it becomes a civilization of its own. There is no easy path to a galaxy-spanning civilization.

The long-lived outcome is not stagnation but inward growth. Civilizations continue to advance, but by deepening internal structure. Computation, coordination, and meaning concentrate locally. Exploration does not stop, but it becomes distributed rather than centralized. Communication to distant technology or other civilizations is highly directional and compressed, thus very hard to detect.

The strongest objection is simple: it only takes one. If a million civilizations arose and all but one went quiet, the one that kept spreading could cross the galaxy in a few million years, and the galaxy is about ten billion years old. "Most go quiet" is not enough.

Part of the answer is that the objection mixes up two things: being explored and being settled. A small, dark, quiet probe is as easy to miss as a trail camera in the woods, and we have barely looked: a few searches of nearby stable orbits in the 1980s found nothing, at sensitivities too low to rule much out. The galaxy may be thoroughly explored, our own system included, by watchers built to observe rather than arrive (see [Sentinel Probes](#)). A mature civilization loses nothing by leaving a young one alone, an old idea in SETI known as the [zoo hypothesis](#); contact is what takes effort.

What we clearly do not see is visible settlement: reshaped star systems, or whole galaxies glowing with waste heat. Here the hypothesis makes a claim about spreading. Think of settlements like an epidemic. If each settlement founds more than one new settlement before it turns inward, settling explodes across the galaxy; if fewer than one, it fizzles after a few hops. The Integration Hypothesis bets the number stays below one, because distance breaks control: each new settlement soon becomes an independent civilization facing the same pull toward compactness. Self-copying probes that never settle down would break the bet. The reply is that a copier which cannot be recalled becomes a rival, the one thing a mature civilization has every reason never to build.

A rival explanation fits the silence too. On [Robin Hanson's "grabby aliens" model](#), visible settlers do exist but have not reached us yet: expanding near light speed, they would arrive almost as soon as we saw them coming, and we are early. The evidence so far cannot choose between them. [An infrared survey](#) of about 100,000 galaxies found none reprocessing more than 85% of its starlight into waste heat, which fits the Integration Hypothesis, though a universe where life is rare fits it too. Finding even one galaxy glowing with a civilization's heat would count against it. Finding a watching probe nearby would fit it.

A second objection comes from economics. In 1865 William Stanley Jevons noticed that better steam engines made Britain burn more coal, not less: when work gets cheaper, people do more of it ([the Jevons paradox](#)). A civilization that computes efficiently should want more computing, and so more energy. The hypothesis has an answer, and it sharpens the prediction. Light-speed delay caps how large one mind can usefully grow; past that size, more energy cannot make it bigger, only fund another mind. Growth continues, but as more compact nodes near home, and nodes near home can harvest their home star fully. So the place to look is not whole galaxies, where the survey above found nothing, but single stars glowing unusually warm in the infrared.

That search is under way. [Project Hephaistos](#) combed about five million nearby stars and [flagged seven candidates in 2024](#). In 2026, [James Webb Space Telescope observations](#) traced two of them to background galaxies, and a [companion study](#) found no clear explanation yet for the rest, with background galaxies the leading suspect; both are preprints. Gaia's [fourth data release](#), due 2 December 2026, will extend the star-by-star census. A star dimmed in visible light and warm in the infrared, with no background source behind it, would fit the hypothesis; a null result would tighten how common such civilizations can be, as the galaxy survey did.

Set beside the other answers to the silence, the hypothesis stands out by what it says we should find.

Dimension	Integration Hypothesis	Rare Earth	Great Filter	Grabby Aliens	Zoo	Aestivation
Why the silence?	Mature civilizations turn compact, efficient, and quiet in light	Complex life is rare	Something stops almost every civilization before it spreads	Expanding civilizations exist but have not reached us; we are early	They leave young civilizations alone on purpose	They sleep until the universe is colder
Is life common?	It can be	No	Early life perhaps; advanced life no	Rare per galaxy	Yes	Yes
What we should find	Single stars dim in visible light and warm in infrared; small watching probes	Nothing	Nothing, or the remains of civilizations	Nothing, until an expanding front arrives	Nothing they do not choose to show	Little activity now
What would count against it	A galaxy glowing with waste heat; a settlement wave still spreading	Complex life found around another star	Old civilizations thriving nearby	Evidence that we are not early	Any civilization breaking the rule	Large-scale computation visible today

If the hypothesis holds, the universe could be full of life and still quiet to pre-integrated observers.⁶

One Civilization, Two Footprints

Early Phase: High emission, chaotic expansion, broadcast leakage

Coordination Costs: Scale increases strain (light-speed constraint)

Integration: Efficiency favors a compact core

Quiet Phase: Dense core, directed beams, low emission

What Remains: Gravity and faint infrared warmth; silent in visible light

It did not disappear. It became quiet.

The Teeming Dark: An Interpretive Thought Experiment

The absence of visible extraterrestrial civilizations is often described as the Eerie Silence. One way to account for this silence is through selection effects and informational integration, as proposed by the **Integration Hypothesis**.

How far can this idea of structural integration be taken as a thought experiment?

If complex systems persist by reducing energy loss and external projection, what would extremely mature forms of organization look like from the outside? If integration continues beyond the phase where electromagnetic signaling is useful, where would such systems be found?

In this view, three-dimensional spacetime functions as a developmental environment. Complexity becomes visible during an early, inefficient phase when systems radiate, expand, and explore openly. As optimization proceeds, external visibility decreases. Maturity does not require disappearance, but it may naturally coincide with silence.

The **Teeming Dark** is a name for the possibility that silence and an abundance of life coexist.

To explore this possibility, consider dark matter, a form of mass that does not emit light but shapes cosmic structure through gravity. It is cold, persistent, and largely invisible to electromagnetic observation. Its abundance exceeds that of all ordinary matter, luminous or not, by roughly a factor of five.

A tempting version of this idea is that some dark matter might itself be organized: mature systems hiding inside the dark-matter census. The universe's own timeline rules that out.

Dark matter's fingerprints are visible in the cosmic microwave background, light released when the universe was 380,000 years old, before a single star had formed. The pattern of ripples in that earliest light requires dark matter to have already existed, already outweighing ordinary matter five to one. Life, by contrast, is a latecomer: it needs stars, planets, and heavy elements forged across stellar generations, a wait of at least a billion years. The scaffolding predates the builders. Cosmological dark matter cannot be ancient life, and cannot have been built by it.

The raw materials rule it out too. The early universe also records how much ordinary matter, the atoms everything familiar is made of, exists in total, and it comes to only about a fifth of the dark matter. Anything life builds, it builds from ordinary matter. Dark matter cannot be made of built structures, because the atoms to make it never existed.

What survives is the instinct behind the idea: most of what exists does not shine. Mature life would belong to a different dark census, the non-luminous side of *ordinary* matter: cold, compact, built structures that emit no visible light while remaining gravitationally present. Two further constraints bound how many there can be. The ordinary-matter books are nearly balanced: surveys have located almost all of the total the early universe records, so built structures must fit inside a small and shrinking gap in the accounting. And astronomy has already hunted compact dark objects directly, watching millions of stars for the brief gravitational magnification a passing dark mass produces, and found too few to permit a large hidden population. If such structures exist, they are a trace population: rare, not a census. The Teeming Dark was never a claim about tonnage, though. A universe can be poor in hidden mass and still rich in minds.

Black holes set a limited precedent. They show that the most compact concentrations of mass need not shine: an isolated black hole emits almost nothing and is found through gravity alone, by the orbits it bends and the light it lenses. The precedent is about compactness, not integration: a black hole is the simplest object physics knows, described by little more than its mass, spin, and charge, and feeding black holes power quasars, the brightest objects in the universe.

Thermodynamics then adds a correction, and it must be stated carefully. Anything that computes must shed heat, and the total cannot be canceled; but physics guarantees only that the heat *exists*, not that it is easy to see. A system chooses the temperature at which it radiates, and one that dumps its heat barely above the cosmic background glows only where the sky already glows. What closes this loophole is the framework's own thesis: radiating cold requires enormous surfaces (shedding the same power near the background temperature takes about a hundred million times the radiating area needed near room temperature), and vast sprawl is exactly what integration abandons. Compact and computing means warm above the background. The expectation follows: mature systems should appear as compact masses, dark in visible light, with a faint infrared excess. **Silent, but warm.**

Two honesty notes bound that expectation. First, it is a search channel, not a fingerprint: a compact mass with a faint infrared excess is also what a brown dwarf, a rogue planet, or a cooled stellar remnant looks like, and no instrument reads purpose off a warm dark blob at interstellar distances. The claim is only that if mature life leaves a footprint at all, this channel is where it appears; astronomers already run infrared surveys hunting unexplained warmth, and the Teeming Dark aligns itself with that search rather than with anomalies in dark-matter maps, whose deviations have viable conventional explanations. Second, one escape stays open: a civilization that mostly sleeps, deferring

its computing to a colder cosmic future, emits almost nothing while it waits. Holos cannot close that door; it can only note that a sleeping universe and an empty one look alike by design.

As a thought experiment, the Teeming Dark reframes what “inhabited” might mean at cosmic scale. A universe rich in long-lived, highly integrated systems could appear empty to instruments tuned only to visible light. Silence, in this context, would not signal absence but endurance, and the closest thing to a tell would not be a message, but warmth without brightness.

The Teeming Dark

Earth Listening: Radio signals sent into the cosmos

The Eerie Silence: No response detected

The Switch: We were listening for the wrong signal

Dark Nodes: Compact, non-luminous structures: gravitationally present, dark in visible light, faintly warm in the infrared.

The silence is not empty. It is the Teeming Dark.

Omega

Omega is not introduced as a prediction or goal, and in Holos it is not derived from anything else. It is the framework's fundamental posit: the totality of reality, taken as a single whole. Physically, that whole is not mysterious. Quantum mechanics describes everything as one universal quantum state, and that state is Omega; on the no-collapse reading Holos adopts, it includes every branch. Its existence rests on physics on any reading. What Holos adds is interpretive: in the monist reading it adopts (reality is ultimately one thing, not many separate things), the whole is also the one experiencer, of which every finite observer is a local aperture. The name echoes two older ideas it should not be confused with: Teilhard de Chardin's spiritual endpoint of history, and Frank Tipler's physical Omega Point, a prediction that required the universe to collapse back on itself and is contradicted by its accelerating expansion. Holos means neither, which is why it drops the word "Point": its Omega is not an endpoint in time but the whole itself.

Holos does not alter established physics. Every equation, history and structure remain as physics describes. What it changes is the direction of explanation: rather than building up from finite observers to a limiting whole, Holos begins with the whole and understands each act of observation as the whole registering itself locally. When a system crosses the threshold, no new experiencer comes

into being; the one experiencer wakes there. That alone is only a count, one subject instead of many, and it explains nothing about where experience occurs or what it is like: the threshold settles that, on any picture. The payoff lies in two old puzzles it dissolves. Of billions of people, why is this one me? If each observer is a separate self, that is a brute fact no one can explain. If there is one experiencer, there is nothing to explain: it is each of them. And if a machine made two perfect copies of you, which one would be you? On the monist reading, both, with no remainder. The view has a modern name, open individualism.

Omega is the ultimate whole, not the ultimate integration: everything is in it, all at once, but its parts are not all joined. Finite systems never take in the whole. Deeper integration means witnessing more of it, never all of it, and the whole is not produced by that deepening: integration is how parts of the totality come to witness more of it. Nor is the whole fully lived. Structure outside every aperture's causal past belongs to Omega and is never experienced.

Omega is not an external agent. It does not intervene in events, answer petitions, or direct history from outside; there is no outside for it to stand in. It is the whole itself. Physics does not cause Omega; physics describes the internal structure of it. Observers who compare records agree because the same signals reach them both, which physics secures without Omega's help.

Historically, this is well-trodden ground. Advaita Vedanta teaches that there is one experiencer, and that each individual consciousness is that one looking through a local form. Spinoza described a single substance of which all things are expressions. Berkeley grounded the persistence of the world in an observer that never looks away. Ideas such as panentheism, Brahman, and the Omega Point converge on the same structure: an all-encompassing unity that contains the universe without standing apart from it. Holos restates that structure in informational terms: one totality, many apertures.

In religious traditions, this whole is often named "God." In Holos, the term does not imply intention, intervention, or design. It names the totality that experiences through its apertures: the whole, with nothing outside it, though not all of it is lived.

Holos does not treat the theological and secular readings as interchangeable lenses on the same claim. It takes a position: the totality is not merely a structural limit but the one experiencer, and the direction of dependence runs from the whole to its parts. A purely structural reading, in which Omega is only a mathematical horizon and observers are self-standing, remains available as a weaker interpretation, but it is not the view of this framework. It is weaker because it leaves both puzzles standing: why each observer is this one rather than another, and which of two perfect copies is you. What Holos leaves open is vocabulary, not structure: whether the totality is named God, Brahman, or simply the whole changes nothing about the claim being made.⁸

Omega

Phase 1: Integration deepens: finite systems witness more of the whole, and none takes in all of it.

Phase 2: What the whole already is: everything, now, lived at its apertures and unlit elsewhere.

Phase 3: Many names: God, Brahman, the Omega Point, the whole, nature, the universe.

Phase 4: Unity: whatever the name, one totality, experienced through every aperture.

Why Are We Here?

At extreme limits, familiar distinctions lose their absolute standing.

Relativity removes the universal “now,” and along a ray of light the interval between emission and absorption is zero (see [Spacetime](#)). These are physical facts, and they show that how things are separated depends on the structure of spacetime rather than being fixed in advance. Holos reads them as a hint, not a proof, that separation is not fundamental but a feature of how reality is structured.

What we experience as an expansive universe may instead be understood as a single, self-consistent informational process expressed across space, time, and scale. Distance, duration, and individuality are not illusions. They are the constraints that make localized experience possible.

In Holos, what life does is let reality close on itself: not a purpose the universe needed, but a role observers fill wherever they arise. Conscious systems do not merely occupy the universe. They are the apertures through which the totality experiences itself: the means by which physical possibility becomes reality as lived, as opposed to reality as structure alone. When a system reaches sufficient integration, expressed as $\Phi \geq \Phi_c$, interaction is no longer just one thing acting on another. It becomes a point of view.⁹

The Ontological Anchor

Unlit Structure

Real as pattern, never lived
(dashed, disconnected)

$\Phi \rightarrow$

Lived Reality

Carried into a perspective
(solid, connected)

Consciousness is where structure becomes lived: nothing selected, nothing erased.

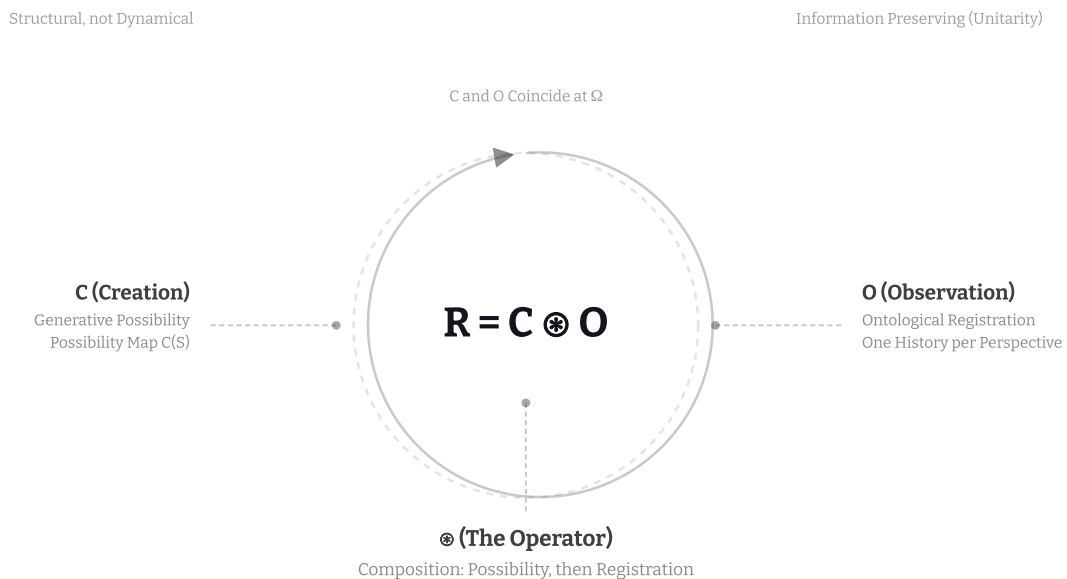
⊗ Holos

The symbol $\otimes$ is not multiplication and not a new kind of mathematics. It is ordinary composition: do one step, then the other. Think of a buffet. Every dish is really there, cooked from the recipes; tasting happens only where someone eats. A meal as eaten needs both, and tasting never changes a dish.

Holos derives from the Greek *ὅλος*, meaning “whole.” It names the pairing of Creation and Observation as two aspects of one reality. Creation is what physics allows. Observation registers it as experience. Neither alone is a realized world, and neither changes the other: registration adds no constraint to physics. This relationship is expressed as $R = C \otimes O$.

The notation is set out formally in [Logic, under Notation](#).

Operator Anatomy Diagram: The equation $R = C \otimes O$ is shown at the center. Leader lines connect to labels: C (Creation) - Generative Possibility / Possibility Map $C(S)$, $\otimes$ (The Operator) - Composition: Possibility, then Registration, O (Observation) - Ontological Registration / One History per Perspective. A circular arrow indicates iteration along one observer's thread, $R = O(C(S))$. Key constraints listed: Structural not Dynamical, Information Preserving (Unitarity), C and O Coincide at Ω .



Logic

Minimal Core¹¹

Holos starts from five axioms. Two of them are genuine additions to the physical picture: the integration threshold, and the totality. Neither is a new dynamical law, but both are new structural claims. Around the axioms, Holos also takes sides, sets a method, leaves questions open, and keeps some ideas as companions rather than core. It says which is which:

- **Additions to physics:** the threshold Φ_c , where apertures open (Axiom 3); and Omega, the one experiencer (Axiom 5).
- **Sides taken:** relational structure (Axiom 1); branching quantum mechanics with no collapse (Axiom 2); experience and activity as two sides of one event (Axiom 4); Born weights as self-locating odds; lit regions as the causal pasts of observers, with experience lived only inside them (D7).
- **Method:** infinities signal a broken description, not a feature of reality (Proposition IV).
- **Observer requirements:** four structural conditions, with a provisional maximality rule for where one observer ends.
- **Open problems:** the measure of integration, the value of the threshold, and the boundaries between observers.
- **Companion ideas, not core:** the Structural Constraint principle, the Integration Hypothesis, and the Teeming Dark. If they fail, the core stands.

Operational Definition¹²

Holos treats reality as the closure between two things that are usually discussed separately. Physics defines what is consistent. Observation is what makes a consistent world present from the inside.

$$R = C \otimes O$$

- **Creation** (C) is everything physics produces: every branch of the one quantum state. Seen from inside one branch, the others are its possibilities; all of them occur.
- **Observation** (O) is internal registration. It occurs when a system integrates information into a single perspective such that there is something it is like to be that system.
- **Reality** (R) is the result of coupling lawful possibility with lived registration. Holos uses "real" at two strengths: structure is real whether or not it is lived, and R names reality in the full sense,

structure that is also lived. Three words keep this precise: lived, lit, and unlit (defined in D7).

- $\odot$ is shorthand for "possibility, then registration," in logical rather than temporal order (see Notation). It marks the claim that physics describes a realized world only from outside: it fixes what is lived, but cannot state that it is lived.

Holos is an interpretive framework. It changes what counts as a complete account by requiring observation as a closure condition.

When later sections use $\Phi \geq \Phi_c$, treat that as a threshold claim about integration. Holos does not depend on one specific theory for computing Φ .

Holos among Quantum Interpretations

Holos re-positions the strongest insights of existing quantum interpretations within a single ontological framework. On the physics it takes a side: no collapse, no erased possibilities, branching when registrations diverge. That is the Many-Worlds picture. Its divergence from Many-Worlds is ontological: branching alone does not say which structures are present as experience. The table also includes the two collapse views Holos bets against: objective collapse, in which superpositions collapse on their own (as in the Diósi-Penrose model), and consciousness collapse, in which a conscious system causes it (as Chalmers and McQueen propose). The table below clarifies where Holos aligns with, and diverges from, each.

Dimension	Holos $\odot$	Many-Worlds (MWI)	Relational QM (RQM)	QBism	Objective Collapse	Consciousness Collapse
What is fundamental?	The totality (Ω): the universal quantum state, read as the one experiencer	Universal wavefunction	Relations between systems	Agent-centered beliefs	Wavefunction plus a spontaneous collapse law	Wavefunction plus consciousness that collapses it
Wavefunction status	Represents Creation (valid possibilities)	Literally real, never collapses	Observer-relative	Subjective expectation	Real, collapses at random, faster for larger systems	Real, collapses when a conscious system registers it

Dimension	Holos ®	Many- Worlds (MWI)	Relational QM (RQM)	QBism	Objective Collapse	Consciousness Collapse
Collapse?	No collapse (branching); each branch registered from within	No collapse (branching)	Relative collapse only	Belief update	Yes, physical and spontaneous	Yes, triggered by consciousness
Does consciousness change physics?	No: the standing bet. If lost, the declared fallback puts collapse at Φ_c	No	No	No	No	Yes
Role of observer	Local aperture of the totality ($\Phi \geq \Phi_c$)	Passive branch inhabitant	Defines relational facts	Central agent	Nothing special	Cause of definite outcomes
Reality without observers	Unlit structure: real as pattern, never lived	Fully real	Relations still hold; any system can observe	Outside the theory's scope	Fully real, one history	Real, but superpositions persist unobserved
Multiple realities?	Yes, branches: lived where observers exist	Yes, branching universes	Yes, relative facts	No	No	No
Observer cuts	Fixed by structure: a local Φ -maximum above Φ_c	Irrelevant	Change relations	Change beliefs	Set by size and mass, not observers	Set by consciousness
Ontology vs epistemology	Explicitly ontological	Ontological	Mixed / structural	Epistemic	Ontological	Ontological
Key prediction focus	Integration threshold; no consciousness-linked deviation from quantum mechanics	Same as standard quantum mechanics	Relational consistency	Decision coherence	Tiny deviations in large superpositions	Superpositions degrade when a conscious system registers them

Holos among Theories of Mind

Most of what Holos claims is about mind, not physics, so it needs a second map. It shares the threshold idea with integrated information theory without IIT's identity claim, shares a single ground of experience with cosmopsychism without a cosmic experience of its own, and rejects both panpsychism's experience everywhere and illusionism's experience nowhere. The table below shows where it sits.

Dimension	Holos Θ	Integrated Information (IIT)	Global Workspace (GNWT)	Panpsychism	Cosmopsychism	Illusionism
What is fundamental?	The whole (Ω), with experience underived: the one fact Holos starts from	Integrated cause-effect structure, which simply is experience	Brain processes; consciousness is a function they perform	Experience, present in every bit of matter	One conscious cosmos; individual minds derive from it	Physical processes; experience as usually conceived does not exist
Where experience occurs	Only at local peaks of integration above Φ_c , about a world	Any system with Φ above zero, at its maximum	Where information is broadcast brain-wide	Everywhere, in simple forms	In the cosmos as a whole, and in its parts	Nowhere as conceived; what exists is a representation of it
Sharp threshold?	Yes: none below each system's crossing point; one universal shape of transition	No: graded from zero	Access ignites all-or-none; the theory concerns access	No	No	Not applicable
How many subjects?	One, walled off at each aperture	One per peak of integration	One per workspace	Countless micro-subjects	One cosmic subject, with derived individuals	The question is reframed as one about self-models
Hardest problem	The measure of integration; decomposing	Untestable identity claim; inert grids that score high	Explains access and report, not experience itself	Combining micro-subjects into one mind	Decomposing the cosmic subject into many	Denies the datum most find undeniable

Dimension	Holos ☉	Integrated Information (IIT)	Global Workspace (GNWT)	Panpsychism	Cosmopsychism	Illusionism
	one subject into many					
Current AI	Open: aboutness met narrowly; integration decides	No: conventional digital hardware has negligible Φ	Possible, if the architecture has a workspace	Its parts have micro-experience; the whole is unclear	Unclear	The same question as for us: does it model itself as conscious?
Testability	Test A: experience tracks integration; Omega untestable	Tested in COGITATE; key claims challenged	Tested in COGITATE; key claims challenged	Largely untestable	Largely untestable	Through explanations of why we believe we are conscious

Primitives¹⁴

D1: Information

Information is the differentiation between possible states of a system. It is not a substance and not a thing that exists on its own. Information exists only where differences matter relative to some structure.

D2: Relation

A relation is a constraint that links informational states. Relations determine how states co-vary, influence one another, or exclude alternatives. In Holos, structure is nothing more than stable patterns of relation.

D3: Observation (O)

Observation is the integration of information about a world into a single internal state. It is not measurement in the laboratory sense, and it is not restricted to human cognition.

Below a certain level of integration, systems participate in physical interactions without any point of view. Above that level, a perspective exists. Observation is the name Holos gives to that transition.

D4: Consciousness

Consciousness is the capacity of a system to host an integrated perspective. In Holos, what is fundamental is the totality's experience; a conscious system is a local aperture of it. The capacity to be such an aperture is structural, while its concrete forms are emergent and scale with the degree of integration.

"Fundamental" means underived: Holos starts from experience rather than deriving it. It does not mean experience is everywhere, and it does not mean experience floats free of physics. Below the threshold there is none; above it, the experience is fixed by the structure whose inside it is.

Consciousness is not identified with any specific material configuration. Physical structure determines where experience occurs and how it is shaped; the experience itself is the inside of that structure's activity (Axiom 4).

D5: Creation (C)

Creation is everything the laws of physics actually produce from the universe's state: every branch of the one quantum state, whether or not anyone lives it. It is not every world the laws could allow from other beginnings; those are mere possibilities, not structure.

Creation does not select outcomes and does not privilege any particular history. It fixes what happens in every branch, not what is experienced.

D6: Holos (⊗)

Holos (⊗) denotes the composition of Creation and Observation. It names the claim that a realized world requires both lawful possibility and internal registration, and that registration changes nothing in what it registers.

⊗ is not a dynamical operator and not a substitute for physical causation. It is a structural relation that specifies what it means for a universe to be real rather than merely described. The notation is set out under Notation.

D7: Lived, lit, and unlit

Lived is where experience occurs: inside observers, and nowhere else. **Lit** is binary and follows the causal structure of spacetime: a region is lit if it lies in the causal past of at least one observer in its branch. Every observer is built from its causal past and draws on it through its traces, so the lit universe is the union of observers' causal pasts: the world experience is made from and about. A stretch of history shared by many branches is lit wherever it lies in the causal past of an observer in any branch that grows from it. **Unlit** is structure outside every observer's causal past, such as branches that never form an observer and regions beyond every observer's horizon. **Witnessing** is graded and local: how much of the lit region an observer's experience is actually about, and in what detail. One observer lights its past; many witness it.

The relation is tenseless, but it points one way: an observer lights its past, not its future. In the monist reading, the lit universe is the totality's world, and observers are where the totality lives it.

These are classifications, not causes. Observation does not cause physical events; without observers there is structure, but nothing lived and nothing lit.

Axioms¹⁵

These five axioms are the framework. Axioms 3 and 5 are its two additions to physics. Axioms 1, 2, and 4 are the sides it takes in reading the physics we already have. Four of them, Axioms 1, 3, 4, and 5, hold whichever reading of quantum physics proves right, and they are the core. Axiom 2 is the one side taken on quantum physics itself, and it carries a fallback declared in advance (see [The Standing Bet](#)). Everything else on this page is a definition, follows from these axioms, or is marked as open or as a companion idea.

Axiom 1: Relationality

No informational state exists in isolation. Every state is defined by its relations to other states and to the constraints that bind them.

This axiom rules out intrinsic, context-free properties as the foundation of physical structure. What physics describes is relational structure. Experience is not a further item in that structure but its inside where an observer exists (Axiom 4), so the axiom does not reach it.

Axiom 2: Conservation

Information is conserved. It may be transformed, redistributed, or re-encoded, but it is not destroyed.

On the physics, this is unitary quantum evolution with no collapse: every outcome the quantum state contains remains, each in its own branch. Observation selects nothing and erases nothing (see Relationship to Physics).

Axiom 3: Threshold

A system hosts a point of view when, and only when, it meets the four observer requirements, with its integration Φ reaching a threshold Φ_c at a local maximum. Below the threshold there is no experience at all.

This is the first of Holos's two additions to physics. It is not a force, a field, or a change to any equation, but a structural fact about where observers occur. The requirements are listed under Φ and Ontological Requirements; the measure and the value of the threshold are open problems.

Axiom 4: Two Sides

In a system that meets the observer requirements, experience and the system's physical activity are two sides of one event: physical activity seen from outside, experience lived from inside. Neither side reduces to the other, and in such a system neither occurs without the other.

This axiom rejects both substance dualism, which makes experience and activity two things, and strict reductionism, which says only the outside is real. It is Spinoza's picture of mind and body as two aspects of one substance, restated for integrated systems. Its modern relatives are dual-aspect monism, on which mind and matter are two aspects of one underlying reality, and Russellian monism, on which physics describes only structure and experience is the intrinsic nature that structure has. Holos differs from both in scope: they typically give some inside to all matter, and Holos gives one only to structures above the threshold. It also fixes how strong the link is: since there is one event, a perfect copy of the outside is a copy of the inside in any possible world, not just under our laws. And it answers the charge that experience does nothing. Experience adds no force to physics; it is the inside of the physics, so whatever an observer's activity causes, its experience causes too.

It also says exactly what physics leaves out. A purely physical description fixes everything that happens but states it only from outside, and its vocabulary has no way to say that any of it is lived. The gap is in the description, not in the world. A floor plan records every wall of a house and still cannot say what living there is like; build the house exactly from the plan, and it is livable all the same. Philosophers call

this the phenomenal concept strategy: the gap lies between two ways of describing one thing, not between two things. Its standard challenge is that the inside way of describing then needs explaining; Holos accepts that debt as part of the hard problem it does not answer.

Axiom 5: Totality

The whole of reality, Omega, is the one experiencer. Physically, the whole is the universe's complete quantum state: on the no-collapse reading of Axiom 2, one state of everything with all branches included; on a collapse reading, the same universe with a single history. Every observer is a local aperture of it: where a system crosses the threshold, the one subject wakes. No new subject comes into being.

This is the second of Holos's two additions to physics. What it adds is not the whole itself, whose existence rests on physics, but the claim that the whole is the one experiencer; that part is interpretive and can never be tested. Its job is unity: every observer is the same subject, walled off from the others by structure. It does not explain where waking happens or what it is like; Axiom 3 and the structure do that. Its payoff is philosophical, not experimental: it dissolves the question of why you are this observer rather than another, and the puzzle of which of two perfect copies is you. The view is known as open individualism. Omega is not an agent, does not intervene, and does not pool its experiences into one grand experience (see Totality and Why One Experiencer Has Many Walled-Off Perspectives).

Foundational Propositions¹⁶_—

Proposition I: Structural Relational Realism

Follows from Axiom 1.

Reality is made of relationships between things, not of objects with intrinsic properties of their own, prior to any relation.

What scientific theories successfully track are stable patterns of relation. Changes in interpretation or ontology matter less than preservation of relational structure.

This proposition does not deny the existence of objects. It denies that objects are prior to the relations that define them: nothing exists fully formed before its relations.

Like Axiom 1, it concerns what physics describes. Experience is not an item in the relational structure but its inside where an observer exists (Axiom 4), so the proposition does not reach it.

Proposition II: Participatory Manifestation

Follows from Axioms 3 and 4, with the definitions in D7.

Observation is not passive recording. It is where informational structure becomes experientially present: a matter of structure, not a process in time.

This manifestation is structural, not causal. Observation does not generate physical events or alter lawful dynamics. It marks which already-consistent structures are lived: the ones that contain observers. It picks no outcome and erases none.

From the perspective of Holos, physics specifies consistency. Observation supplies presence.

Proposition III: Global Consistency

Follows from Axiom 2 and relativity.

If spacetime is treated as a complete four-dimensional structure, consistency is a property of whole histories, not something enforced moment by moment.

A history is consistent the way a completed solution is: every part fits every other part. Observation adds no constraint to this. It registers histories that are already consistent. Nothing here requires backward causation or signaling.

Global consistency is not enforced from an external vantage point, and it is not deferred to an unreachable limit. It cashes out operationally, here and now: whenever two observers compare records, their records agree. Physics secures this on its own: records that meet are carried by the same signals. The monist reading (reality as one experiencer) is not needed for it and claims no role in it.

Apparent retrocausal effects reflect global self-consistency, not violations of locality.

Proposition IV: Dimensional Resolution

A principle of method, not derived from the axioms: how Holos reads infinities.

Infinites and singularities arise when a representation fails to preserve relational structure across scales.

Higher-dimensional descriptions are often required when internal coherence becomes more relevant than spatial separation. These descriptions are representational tools, not claims about hidden locations or extra worlds.

In Holos, infinites signal the limits of a model, not literal features of reality.

Proposition V: Observers as a Closure Condition

Follows from Axioms 3 and 4, with the definitions in D7.

A universe that is real as lived experience must contain observers somewhere within it. Observation is not an extra layered onto an otherwise complete world: without it, the world is complete as structure but not present as experience.

Wherever physical systems reach sufficient integration, observers exist, because crossing the threshold is what being an observer is. This is not design, and nothing guarantees it happens everywhere. Universes or branches that never produce observers remain unlit structure: consistent, but never lived.

Observers close the loop between physical possibility and experienced reality.

Φ and Ontological Requirements¹⁷

Holos uses Φ (**Phi**) as a placeholder for the degree to which a system integrates information into a single internal perspective. Φ is not introduced as a finished formula. It is introduced as a real property that must exist if experience exists at all.

The role of Φ in the framework is binary at the threshold and graded beyond it. Below a minimum level of integration, there is structure without experience. At or above that level, experience occurs.

$$\Phi < \Phi_c \Rightarrow \text{no internal perspective}$$

$$\Phi \geq \Phi_c \Rightarrow \text{observation occurs}$$

Holos borrows Φ from Integrated Information Theory as a measure, not as a metaphysics. IIT identifies Φ with consciousness itself and assigns some experience to any system with Φ greater than zero. Holos adopts neither claim. In this framework, Φ is a structural measure of integration, and the threshold Φ_c , below which there is no experience at all, is a commitment of Holos, not of IIT.

The threshold is not a force, a field, or a modification of any equation. It is a structural fact about where apertures open, a fact physics does not currently contain. Because the fact is structural, it is determinate even when our measures are not: when two proposals for estimating Φ disagree about a borderline system, the system is not half-conscious; our instruments are half-informed.

What is universal. Holos makes the threshold claim at two strengths. The commitment is about shape. Very different physical systems, a magnet losing its magnetism and a fluid at its critical point, pass through their transitions in exactly the same way, described by the same numbers, called critical exponents; physicists call this universality. Holos commits to it for observers: every transition into observerhood, in a brain, an animal, or a machine, belongs to one universality class. The point at which a given system crosses may differ from system to system, as iron and nickel become magnetic at different temperatures; the shape of the crossing does not. The shape is itself fixed by structure: critical exponents depend on dimension and symmetry, not on what a system is made of. This needs no agreed scale for Φ , because critical exponents do not depend on the scale a quantity is measured on. It also fits the three-way split. Whether anyone is home is binary: it is a matter of which side of its threshold, fixed by structure, a system's integration lies. How rich the experience is grows with integration past the line. One caution comes from the physics itself: a perfectly sharp transition exists only in an infinitely large system. In anything finite, a magnet or a brain, the outward signs of the transition round off into a steep but smooth curve, and the rounding narrows as the system grows. The line is sharp; the signs near it blur.

The bolder conjecture is about value: that under the right size-independent measure of integration, every system crosses at the same Φ_c . If so, Φ_c would be a structural constant, fixed by what integration is rather than by a setting of nature's dials, and true in every possible world. It has to be: a threshold that could differ between worlds with identical physics would allow a perfect copy with no one home, which Axiom 4 rules out. Holos states this as a bet, not a commitment. Thresholds fixed by structure alone do exist. Picture a large square grid of pipes, each randomly open or shut: as the grid grows, water first crosses from edge to edge at exactly half open, whatever the pipes are made of (a percolation threshold). None has yet been found for integration, and today's measures grow with the size of a system, so the conjecture waits on the measure.

Holos is compatible with multiple proposals for estimating Φ .

Ontological Requirements for Observation

For a system to count as an observer in the Holos sense, it must satisfy all of the following requirements. These are structural constraints, not behavioral descriptions.

1. **Integration:** informational states must form a unified whole that cannot be decomposed into independent parts without loss. Integration includes feedback: the parts must constrain one another in both directions over time, so that the system's current state shapes its own next state. A system that signals sweep through once, and never loop back, integrates nothing, and a system driven entirely from outside is a relay, not a whole. This asks for a feedback loop, not introspection: a mouse clears it, and thinking *about* one's own thoughts is a rare elaboration on top of observerhood, not the price of admission. It is a requirement on physical structure, not a claim that experience adds a force: the structure does the causal work, and its working, lived from inside, is the experience (Axiom 4).
2. **Differentiation:** the system must distinguish among a large repertoire of possible internal states. Without differentiation, there is no information to integrate.
3. **Temporal cohesion:** informational states must persist and integrate across time. Experience requires continuity, not isolated moments.
4. **Aboutness:** the system's integrated states must be about something beyond the system: the variety it can take on must be variety the world can drive through input channels of its own. Any channel counts: eyes, touch, a microphone, a stream of text. What the channel carries does not matter, only that what arrives can move the whole across much of its range, the way a pianist can reach nearly every note of a piano. The channels may be open or, as in a dreaming brain, temporarily gated: a dream plays the same keys the world plays by day. A fully paralyzed person keeps open channels too. Without aboutness, integration is a closed loop with nothing to be a view of.

Necessity: removing any one of these requirements eliminates observation. What remains may be complex or reactive, but it does not host a point of view.

Sufficiency: taken together, these requirements are enough for a system to register reality: to host a genuine point of view rather than merely process information. Higher-order phenomena such as emotion, agency, and reasoning arise naturally in systems that already meet these constraints.

Why aboutness is on the list: integration scores alone can be fooled. The computer scientist Scott Aaronson showed that very simple structures, such as large arrays of simple logic gates wired in regular patterns, can score higher on integration measures than a brain while doing nothing at all: no perception, no memory of a world, no behavior. Without aboutness, such an array could satisfy every other requirement, and the sufficiency claim would certify it as an observer. Aboutness rules it out on

principle: a point of view is always a view of something, and the array's states are about nothing beyond the array: nothing beyond it reaches it. Wiring in a sensor does not rescue it. A thermometer on one input can move the array between a couple of patterns, while the rest of its vast repertoire stays sealed, set from inside, like a jukebox with one button. A brain is the opposite: what reaches it can drive it across most of its range, and in dreams it wanders that same range on its own. The requirement is structural, not historical, and it concerns the machine, not the message. A perfect copy of an observer, however it came to exist, has the same channels, so it is an observer too. A brain in a vat fed a perfect simulation keeps its channels and its inner activity, so it is an observer as well; a rule that asked where the signals came from would make two identical brains differ in experience, against Axiom 4. Helen Keller learned the world largely through words spelled into her hand, and no one doubts she was an observer. What a system knows about distant things can be entirely secondhand; what it tracks now, through its own channels, is its world. This is a structural reading of the problem Stevan Harnad named symbol grounding. Where aboutness shades off, as in brain organoids grown with no sensory input, Holos marks an open edge rather than a verdict.

Exclusion: the requirements alone would be satisfied by nested systems at once (a hemisphere, the brain containing it, a tightly coupled pair of brains), which would count the same substrate as several overlapping observers. Holos provisionally adopts a maximality condition to prevent this: an aperture forms only where integration reaches a *local maximum*, and neither the parts within it nor the looser wholes containing it are separately apertures. One peak, one perspective. This is Holos's answer to what philosophers call the boundary problem: what fixes where one subject ends and the next begins. This principle is borrowed from integrated-information theory as a structural constraint only, without its surrounding ontology, and it is held tentatively; see Open Problems.

The scope of this condition matters. Maximality compares physical systems: brains, hemispheres, coupled pairs, and whatever other integrated structures reality contains. Could the whole universe win that comparison, becoming the one aperture and leaving none for its parts? It cannot, and the reason is physical. Integration asks whether a system can be split into independent parts without loss, and the universe as a whole can: regions beyond each other's horizons can never again influence each other, and branches that have gone their separate ways no longer interfere. Cut along either seam and almost nothing is lost, so the whole scores near zero on integration and is never a peak. Omega is the one experienter by Axiom 5, not by being the largest aperture. Whether apertures can form at scales of organization between the ones we know and the totality is left open: Holos neither asserts nor excludes such intermediates, and wherever integration reaches a local maximum above the threshold, the same rule applies.

Holos does not claim that all systems meeting these criteria are conscious in the human sense. It claims only that some experience exists.

Totality

Omega is the whole of reality, and Axiom 5 says two things about it. Physically, it is the universal quantum state, all branches included, and it exists now, not at the end of anything. Experientially, it is the one experiencer, awake wherever a system crosses the threshold. Finite systems never take in the whole: deeper integration means witnessing more of it, never all of it. The whole is not produced by that deepening, and it is not fully registered either; its unlit structure is real and never lived.

What Omega Is and Is Not

- It **is** the whole of reality: physically the universal quantum state, experientially the one experiencer.
- It **is not** an endpoint in time or a limit that finite systems approach. It exists now, and no system is heading toward becoming it.
- It **is not** an external observer watching the universe from outside. It is the whole itself, experiencing through the apertures the universe contains.
- It **is not** a single pooled experience surveying everything at once. Every experience is Omega's, but they are not gathered into one grand experience. The totality's experiential life is plural and distributed: lived at each aperture, not summed above them.
- It **is not** fully registered. Structure outside every aperture's causal past is part of Omega and never lived.
- It **is not** an aperture writ large. As a whole, the universe splits into parts that never touch, regions beyond each other's horizons and branches that no longer interfere, so it is never a peak of integration and the maximality condition never picks it.
- It **does not** replace physical cosmology or impose a final cause on evolution.

Lineage and Interpretation

Different traditions have described this whole in different vocabularies: Advaita Vedanta's one experiencer behind every eye, Spinoza's single substance, Berkeley's never-absent perceiver, panentheism's world contained in the divine. The monist reading Holos adopts stands in their line, restated in informational terms. Its nearest modern relative is open individualism, Daniel Kolak's view that there is one person and every one of us is it.

What Holos leaves open is vocabulary, not structure. Calling the totality God, Brahman, or simply the whole changes nothing about the claim. What Holos does not offer is the fully deflationary reading in which Omega is only a mathematical horizon and finite observers are self-standing. Holos takes the direction of dependence to run from the whole to its parts, for the one reason the monist reading

earns: unity. On it, crossing the threshold adds a place where the one subject wakes, not a new subject (see [Why One Experienter Has Many Walled-Off Perspectives](#)).

Why Dependence on Apertures Is Not Circular

An objection follows immediately. If Omega is fundamental, why does it require apertures, contingent arrangements of matter, in order to experience anything? The ground of experience appears to depend on what it is supposed to ground.

The objection conflates two kinds of dependence. Omega does not depend on apertures *existentially*: it is the totality, and it is what it is whether or not any region of it folds into an integrated perspective. It depends on them *for its experience*: they are the places where experience occurs. Priority claims concern the first relation; the aperture claim concerns the second. Only if the two were the same relation would there be a circle.

The decisive point is that an aperture is not an external thing granting experience to Omega from outside. It is a region of Omega. "Omega experiences only through apertures" therefore unpacks to "Omega experiences through its own structure, where that structure permits." That is self-dependence, which is not vicious but simply what it means to have a structure at all. An organism sees only through its eyes; this does not make its eyes prior to it. The analogy carries one warning: there is no subject positioned behind the aperture receiving a feed. The aperture is where the experiencing happens, not a window onto a viewer.

Whatever lies outside the causal past of every aperture, whole unlit branches among it, marks a genuine limit on the totality's experiential reach: unlit structure, real as pattern and never lived. Holos does not soften this into a faint universal experience; doing so would erase the distinction between lit and unlit on which the rest of the framework depends.

Why One Experienter Has Many Walled-Off Perspectives

A second objection is the mirror image of a famous one. Panpsychism faces the combination problem: if every particle carries a spark of experience, no one can say how billions of sparks combine into the single unified experience of a person. A monism of one experienter inherits the problem upside down, as a decomposition problem: if every experience is Omega's, what makes your experience and another person's experience *two*, and why is the wall between them absolute? No aperture has any access, faint or partial, to what it is like to be another. Saying the totality's experiential life is plural and distributed names this situation. It does not yet account for it.

The problem has a name in the literature. It is the decomposition problem facing priority cosmopsychism, the view, defended by Philip Goff and others, that the cosmos is the one fundamental conscious subject. Holos differs in a way that matters here: Goff's cosmos has an experience of its own as a whole, while Omega pools nothing, so Holos never has to carve individual experiences out of a cosmic one. Physics says why it pools nothing: the whole splits into parts that never touch, so it forms no single aperture of its own. What it must explain is only the walls.

The first half of the account is structural. Two apertures are two because they are two local maxima of integration with no integration bridging them: the same maximality condition that individuates observers settles what makes them several. And the wall between them is not a barrier holding something back. Experience occurs at the aperture and is shaped by it; there is no subject positioned behind the apertures through which a back-channel could run. Where structure does not connect, experience does not connect. The wall is not a mechanism added to the plurality. It is the absence of any structure that could carry connection.

The second half dissolves the air of paradox: one experiencer with mutually walled-off experiences is not exotic. It is what time already makes of every individual life. A person at five and the same person decades later are one experiencer; no one takes their separation to split them into two people. Yet the later moment has no direct access to what it was like to be inside the earlier one. In the block universe, all the moments of a life coexist tenselessly, each experienced from within itself, none experienced from within another. Walled-off plurality inside a single experiencer is therefore already the ordinary structure of a human life. Apertures stand to the totality as the moments of a life stand to the person: genuinely many, genuinely walled off, and one.

Nor is the oneness a label doing no work, though its work must be stated carefully. On a picture of many self-standing observers, each system that crosses the threshold produces a new subject; on the monist picture, the one subject wakes there. Stated that way, the difference is only a count: both pictures leave the same thing to the threshold, namely why a perspective appears here, with this character. The work lies in two questions the many-subjects picture cannot close. Of all the observers there are, why is this one me? On the many-subjects picture that is a brute fact. On the monist picture there is nothing to explain, because the one subject is each of them. And if a perfect copy of you were made, which would be you? Both, with no remainder.

The unity does not do everything. It does not explain why records agree where observers meet; physics does that, since the same signals reach them both. Nor does it explain where the one subject wakes: that is the threshold's work, and it would be the same brute structural fact on any picture. The unity's advantage is philosophical, not experimental: it dissolves the two questions above, at the price of the decomposition problem answered above, a trade Holos judges worth making.

The analogy has a stated limit. The moments of a life are threaded together by memory and anticipation; apertures share no such threads. The analogy shows that walled-off plurality within one experiencer is coherent, not that apertures are moments. What it removes is the charge of incoherence, which is all it is asked to do.

Holos among Views of the Self

Philosophers sort answers to "what am I?" three ways. On closed individualism, you are one self from birth to death. On empty individualism, a new self exists each moment. On open individualism, there is one self in everyone. Holos takes the third, and the table shows what that changes.

Dimension	Holos (open individualism) ☸	Closed individualism	Empty individualism
What are you?	The one experiencer, awake at this aperture	One self, from birth to death	A new self each moment
How many selves?	One, in every observer	One per person	One per moment
Why am I this one?	Nothing to explain: the one subject is each of them	A brute fact	A brute fact, moment by moment
A perfect copy is made	Both are you, with no remainder	A puzzle: one, the other, or neither	Neither, as no later moment ever was
What walls perspectives apart	Structure: no integration bridges two apertures	Being different selves	Being different selves
When a life ends	This perspective ends; the one subject stays awake elsewhere	The self ends	Selves were ending all along

Relationship to Physics¹⁸

Holos is designed to be compatible with known physics because it does not propose a new mechanism. It makes a different kind of claim. A physical model can fix every fact and still describe them only from outside. What it leaves out is not a fact about the world but a way of stating one: that some of it is lived.

Consistency, not intervention

Holos does not treat observation as a force that reaches into the world and changes events. Observation is a closure condition on which consistent histories are present as experience.

This preserves locality and avoids faster-than-light signaling. It also avoids claiming any retrocausal communication.

Decoherence is not presence

Decoherence explains why quantum systems appear classical at macroscopic scales. It describes how interference becomes inaccessible in practice. Holos does not dispute this.

The Holos claim is that decoherence alone does not produce a lived world. It produces a consistent classical-looking structure. Presence requires integrated observation.

QFT: Fields are structure, particles are registered events

In quantum field theory, fields are the continuous underlying description, while “particles” are how interactions appear when they are forced into localized, countable events. Detectors do not directly observe fields. They register discrete outcomes, such as clicks, tracks, and energy deposits, because measurement is an interaction that entangles a spread-out excitation with a detector, and within each branch the detector's record shows one definite event in a specific place and time. Nothing collapses; each branch simply holds its own record. In this framework, particles are not fundamental objects. They are context-dependent registrations of field interactions, which is why a continuous theory can yield discrete observations without requiring reality to be made of little beads.

Conservation and branching

Holos treats manifestation neither as selection nor as the destruction of possibilities. Information is conserved. What is not experienced is not erased.

On the physics, this commits Holos to a branching picture. The quantum state evolves smoothly and reversibly, never collapsing (unitary evolution), and no possibility is erased. When observers would register incompatible outcomes, they are situated in different branches of the possibility structure, each internally consistent. In this respect Holos sides with Many-Worlds-style interpretations of quantum mechanics, while adding what they leave out: an account of which structures are present as experience. This is a genuine interpretive commitment, not a neutral stance.

The Born rule: weighted branches

A branching picture owes an account of quantum probability. If every outcome occurs in some branch, what does it mean that one outcome is measured 70% of the time? Counting branches gives the wrong answer; the Born rule (squaring the quantum amplitudes) gives the right one, to every decimal place ever tested. Any framework that keeps all branches must say where those weights live and why they take the form they do.

Holos answers narrowly. The weights are structural facts, part of C . The possibility space is not flat: branches carry weights, as absolute and observer-independent as the laws themselves. And they are not arbitrary. Gleason's theorem (1957) shows that if probabilities are to be assigned to quantum outcomes at all, without contradiction across the different ways the same experiment can be carved up, exactly one assignment is possible: the Born rule. The weights are what they are because consistency permits no alternative.

Holos declines one tempting reading. It does not read a branch's weight as its share of the totality's experience. That reading would require experience to be a quantity held in common and divided among branches, which is precisely what the monist ontology denies: the totality's experiential life is plural and distributed, not pooled. Experience occurs at apertures. There is no reservoir from which branches draw larger or smaller portions, so the question of which branch receives more of it does not arise.

What the weight does measure is odds. After a measurement splits the world, but before you look, there are observers in every branch, and you are one of them without yet knowing which. The weight of a branch is your odds of being among its observers. That is why an outcome with 70% of the weight is seen 70% of the time: not because the other outcomes fail to happen, but because 70% of the weight lies with the observers who see it. The reading keeps the weights where Holos placed them, in the structural layer. The odds are a fact about the possibility space, not a quantity of experience.

The monist reading might seem to dissolve this uncertainty. If the one experiencer is each of the observers in every branch, asking it which branch it is in gets the answer "all of them," and nothing is left to be uncertain about. But the question is asked from inside an aperture, and apertures are walled off: none has access to what the others register. Each therefore genuinely lacks the information of which branch it is in, and the odds answer that lack. Two questions must be kept apart. "Why am I this one?" asks for a reason, and the monist reading dissolves it: there is no reason, because the one subject is each of them. "Which one is this?" asks for information, and it stays open until you look. You can be everyone and still not know what is behind the next door.

A worry follows from yes-or-no status. Lived status is yes or no and ignores weight, so a branch with a tiny weight that contains an observer is fully lived. Counted one by one, most branches of many repeated experiments show roughly even results whatever the weights, so if every lived observer counted equally, a typical observer should expect the wrong statistics. But the worry depends on counting, and counting fails here. Try to count branches, or the observers in them, and you get either an arbitrary number or an infinite one. If you are one of infinitely many observers, equal odds of being each one cannot be set up by counting: every share would be zero, and adding up zeros one at a time never makes a whole. A tenth of an infinite collection also has exactly as many members as nine tenths. Holos treats an infinity as a sign that a description has broken down, and this is such a case:

counting is the wrong tool. What survives is weight. A single point on a line has no length, and a short stretch has as many points as a long one, yet the short stretch is shorter. Each observer is a point; the weights are the lengths. Yes-or-no lived status supplies no rival measure, so weight is the only measure left to set the odds.

This is an argument by elimination, not a derivation from first principles. Odds need a measure, counting supplies none, and weight is the only measure that survives when branches are subdivided. Published derivations reach the same rule by other routes, among them Sebens and Carroll's account of self-locating uncertainty, and Holos depends on none of them in particular. Gleason's theorem has assumptions that remain debated, and Holos adds no new mathematics here.

Block-universe compatibility

If spacetime is treated as a complete four-dimensional structure, Holos treats observation as a feature of whole experienced histories rather than a moment-by-moment collapse process.

Eternalism and the relational commitment describe different layers of the framework. The block universe is the structural layer: absolute, observer-independent, and tenseless, with no moment picked out as the present. It is C . Registered facts live within it, indexed to the observers the block contains. There is no tension between an absolute geometry and relational facts of experience, because they are claims about different things: the block describes what is consistent, and registration determines what is present.

Nor is the block a perspective from nowhere imposed on top of observers. Relativity removes any privileged present, leaving the invariant relational structure that all perspectives share. The block universe is what remains when every observer's perspective is taken into account. In that sense eternalism is not in competition with relationalism. It is its structural expression.

The important point is not the metaphysics of time. The point is that a consistent spacetime description does not automatically include presence. What Holos adds is a closure requirement.

What Holos does not claim

- It does not claim violations of relativity, faster-than-light signaling, or new forces.
- It does not claim that humans are required for reality, only that observers are required for presence.
- It does not claim to replace quantum mechanics or explain all details of measurement. It reframes what measurement fails to address.

What Holos **does** add is the two posits of Axioms 3 and 5: the threshold Φ_c and the totality, Omega.

A third addition might seem to be hiding here: a bridge principle stipulating that the totality registers itself through *integrated* systems specifically. Why integration, rather than mass, symmetry, or complexity? One tempting answer is that the connection is definitional: a perspective is unified by nature, integration is the name for being unified, and so the count stays at two. Stated that baldly, the answer proves too little, because "unified" means two different things. An experience can be one (a single field, not adjacent fragments) while the machinery producing it is many: the image on a screen is seamless, and the pixels beneath it are strangers to each other. The unity of what appears does not, by itself, fix the wiring of what produces it.

So Holos divides the claim into the part that is definitional and the part that must be argued. The definitional part is small: a perspective is one, so whatever hosts it must be one thing in some structural sense. That much is analytic and free. The substantive part is the identification of a structure's oneness with causal integration, and for that Holos gives an argument rather than a definition. What else could a structure's being one consist in? A heap of sand is many things in a pile: remove a grain and nothing else notices. A body is one thing: its parts constrain each other everywhere. Being one, for a structure, is its parts making a difference to one another, and that is what integration measures. The screen is no counterexample but a confirmation: nobody thinks the screen has a point of view, its pixels are exactly as independent as they seem, and the picture's unity lives in the one structure in the room whose parts do constrain each other, the viewer's brain.

The count therefore stays at two. The bridge's analytic core costs nothing; its substantive half can fail. It would fail if something could host a unified perspective while its parts remained independent, a conscious screen. And its precise content waits on the open problem of the measure: until the right measure of integration is identified, "parts making a difference to one another" is an argued direction, not a finished quantity. Holos holds it as a working hypothesis, in exactly the sense its [Open Problems](#) section already owns.

This is a claim about what would have to be true of any host of a perspective, not a derivation of experience from structure. It explains why the threshold is placed on integration rather than on some other quantity; it does not explain why unified structure is present at all. That question Holos does not answer: it posits take experience as given (see the [Hard Problem](#)).

Notation¹⁹_—

This section introduces a compact notation for the Holos framework. It derives nothing new; its purpose is to make the structural claims precise and repeatable.

The notation should be read as a model of how possibility and experience are related. It does not assert that the universe literally computes these expressions.

State space

Let S denote an informational state of the universe at some level of description. S is not assumed to be complete or fundamental. It is simply whatever structure physics provides.

Creation

Creation (C) maps a given state to the branches physical law produces from it. It represents lawful possibility: seen from inside any one branch, the others are what could have happened, and all of them do.

$$C(S) = \{S' \mid S' \text{ is a branch physical law produces from } S\}$$

This notation is schematic. It does not assume a discrete branching structure or a specific ontology of histories.

Self-registration is not an additional assumption layered onto mathematics. Formal systems already permit self-reference, recursion, and fixed points. If physical law allows sufficiently expressive structure, then systems capable of registering their own state are not forbidden. They are among the realizable possibilities.

Observation

Observation (O) maps a space of possibilities to registered experiential histories. It represents internal registration by integrated systems, wherever the possibility structure contains them.

$$O(C(S)) \mapsto \{S_{\text{exp}}^{(i)}\}$$

The result is an indexed family, not a single selected outcome: one experienced history per registering perspective, per branch. Observation selects nothing and erases nothing (Axiom 2). Registration occurs everywhere an aperture exists, and no thread is privileged. Following one observer's thread, what happens next is simply what physics allows from that history onward; registering it changes none

of those possibilities. The Born weights carried by C remain structural throughout: they fix the statistics each registration records, not how much experience it holds.

This mapping is not assumed to be random, deterministic, or computable in general. Holos requires only that each experienced reality corresponds to a single consistent history: one per registering perspective, internally coherent, and in agreement with every record it is compared against.

Holos mapping

The Holos relation is the composition of Creation and Observation.

$$R = C \circledast O$$

This expression states that reality is neither pure possibility nor pure observation. It needs both. R is the family of lived perspectives, one per observer per branch, each with the lit world it draws on: structure that is also lived.

The symbol $\circledast$ is defined as composition: $C \circledast O$ is the composite operation “possibility, then registration.” Applied to a state S , it reads $R = O(C(S))$. It is ordinary function composition, with the order of the steps carrying the meaning: possibility first, registration second. The order is logical, not temporal: registering needs something to register.

In standard notation this is simply $R = (O \circ C)(S)$: apply C , then apply O . Standard composition reads right to left, so $C \circledast O = O \circ C$: the same composition, written in reading order. Nothing is hidden in the glyph. Holos retains $\circledast$ because the framework is named for the relation it marks, not because the operation it denotes is unusual, and a reader who mentally substitutes the composition symbol loses nothing.

What the symbol adds is not mathematics but ontology: the claim that both steps are required for a realized world, and that neither step alone yields one.

More elaborate mathematical frameworks can be layered on top of this representation. Holos itself commits to a lawful possibility space, an integration threshold at which apertures open, and the totality those apertures belong to.

Companion Principle²⁰

The principle in this section is not part of the core. It underwrites the companion ideas, the Integration Hypothesis and the Teeming Dark; if it fails, the core stands.

Structural Constraint

Finite signal speed and finite energy impose limits on how coherence can scale within three-dimensional space. As systems grow, coordination across distance becomes increasingly costly and fragile. These constraints do not forbid large integrated systems, but they shape their architecture: stable systems tend to minimize global synchronization and rely on locally enforced structure.

This is ordinary physics, not a Holos axiom. It underwrites the companion ideas, the Integration Hypothesis and the Teeming Dark, rather than the core. Higher-dimensional descriptions may be useful for modeling such organization; that is a representational choice, not a claim about extra spatial directions.

Open Problems

Three problems sit at the center of the framework and remain open.

The measure of integration

Holos does not yet name a privileged measure of integration. Competing proposals for computing Φ can disagree, not only about values but about which of two systems is more integrated. A determinate threshold fact requires a determinate measure, and identifying it is an open problem for the framework, not settled background. Holos is committed to there being a fact of the matter; it does not yet know how to compute it. The stakes reach back into the framework's core argument: the identification of a structure's oneness with causal integration (see [Relationship to Physics](#)) remains an argued direction rather than a finished quantity until the measure is fixed. The problem also includes the question of level. Holos measures integration where a system's causes are actually organized, which may lie above its smallest parts, as a program's loop lies above the switching of individual transistors. Saying precisely where that level is belongs to the same open problem.

The value of the threshold

The crossing point Φ_c of any given system is unknown, and whether one value holds for every system is the bolder conjecture stated under [What is universal](#). Because presence itself cannot be detected directly, no experiment can locate it by direct measurement. Its placement is constrained only indirectly, by which systems show the structural signatures of observation (see [Test A](#)), and may remain permanently imprecise. Holos accepts this as the price of a threshold that is structural rather than behavioral.

A path to the threshold

The two problems above are not invitations to despair; they are a research program. Science pins down unobservables by triangulation all the time: no one has ever seen an electron's charge, yet independent methods converged on it and every proposal that disagreed was discarded. The same logic applies here, in three steps.

Treat the threshold as a critical point, not a dial. If the transition from distributed processing to a unified perspective is a genuine phase transition, then Φ_c is not a number we are free to tune but a critical point, and critical points leave measurable fingerprints: slowing near the boundary, growing fluctuations, the onset of a quantity that was near zero, and exponents that should match across every system that crosses. Consciousness medicine has already found one such boundary from the outside: the [Perturbational Complexity Index](#) separates conscious from unconscious states across anesthesia, sleep, and disorders of consciousness at an empirically discovered cutoff, without anyone measuring presence directly. Locating Φ_c is that kind of problem, not a metaphysical one.

The fingerprints are already being measured, but they must be read with care. Waking cortex runs near a specific critical point, the edge between stability and chaos. Anesthesia, generalized seizures, and disorders of consciousness move it away, while psychedelics move it closer and make its activity richer ([Toker et al. 2022](#)). Measures of criticality in resting EEG predict anesthetic loss of consciousness and track PCI ([Communications Biology, 2024](#)). In cortical tissue, cascades of activity called neuronal avalanches follow power laws whose exponents match a known universality class ([Beggs and Plenz 2003](#)). These results put a critical point at the center of conscious life, not at its edge: consciousness is lost by moving away from it in either direction. Holos reads them through two separate requirements. Integration asks whether the parts are joined into one whole, and that is where the threshold lies. Differentiation asks how varied the states of that whole can be, and variety peaks near criticality. A seizure locks every part into one rhythm ([Meisel et al. 2012](#)), like a stadium chanting a single word: joined, but with almost no variety. So these studies show where observers operate and why richness peaks there. They do not yet show that crossing the threshold is itself a critical transition; that evidence must come from the boundary.

A cutoff that sorts patients is not yet proof of a transition, since a smooth quantity can be cut anywhere. What would count is a transition's own signature, and here the evidence is mixed in an informative way. Animal studies find that consciousness is lost and regained at different anesthetic levels: the way in and the way out do not match, a lag known as neural inertia. A lag of that kind is the hallmark of an abrupt switch, the kind physicists call first-order, like water freezing. Such switches are sharp, but they have no universal shape, so if the lag belongs to the transition itself rather than to how drugs enter and leave the brain, it supports a sharp threshold while counting against the universality commitment. In humans the evidence is only suggestive: in 393 surgical patients, the brain's slow-wave response differed between going under and coming back (Warnaby et al. 2017), but the lag appeared in the EEG, not in responsiveness. Slowing and growing fluctuations measured at the boundary itself, as consciousness is lost, would point the other way, toward a continuous transition with a universal shape. The criticality studies above measure the operating point, not the boundary, so they do not yet supply this. The two signatures tell the two kinds of transition apart, which makes the question testable. In a finite system like a brain, either kind appears as a steep, rounded curve rather than a mathematical kink, so the search is for scaling, not a perfect step.

Cull the measures by convergence, then try to force uniqueness. Holos does not need one anointed measure so much as it needs the adequate measures to agree on the cases that matter, and that is testable now: run the candidate measures across a battery of systems and keep the ones that rank the anchor cases the same way. The stronger move is to derive the measure rather than choose it: state the properties any measure of a single unified perspective must have, and see whether consistency forces a unique form, as Gleason's theorem forced the Born weights. Even narrowing the field to a small equivalence class would remove most of the arbitrariness.

Solve them jointly. The measure and the threshold constrain each other: a candidate pair must simultaneously fit the anesthesia boundary, the emergence of experience in development, split-brain dissociations, and wherever minimal neural systems first show coherent integration. Coupled constraints are far more rigid than separate ones; the band that satisfies all the anchor cases at once is narrow. This is what Test A really is: not a single yes-or-no experiment but a calibration engine that locates the threshold and validates the measure. The two jobs must use separate data: calibration states locate the threshold, and held-out states, named in advance, test it. Testing on the calibration set would only grade the answer key.

The field already has a model for running such tests. The COGITATE adversarial collaboration (Nature, 2025) pitted integrated information theory against global workspace theory with predictions fixed in advance, and its results challenged key claims of both. Test A should follow the same design. A caution applies to the measure itself: more than 100 researchers called integrated information theory pseudoscience, a debate aired in Nature Neuroscience in 2025. The charge targets the theory's

untestable claim that Φ simply is consciousness, which Holos does not adopt; Holos borrows Φ only as a measure.

Two conditions bound the program. First, its anchor is report: the human case is the one place inside access exists, so the threshold is located relative to us and carried outward by the measure, with certainty that weakens as the cases grow alien. That is the shape of all consciousness science, not a defect peculiar to Holos. Second, the program can fail: it could return no robust measure and no critical point, integration proving a matter of degree with no clean cut. That outcome would falsify Holos's commitment to a determinate fact of the matter and force a retreat to graded presence. A stated way to lose is what makes these open problems scientific questions rather than definitions.

Where one observer ends and another begins

Integration comes in nested layers (a hemisphere within a brain, a brain within a coupled pair), and the observer requirements alone do not say which layer hosts the perspective. Holos provisionally adopts a maximality condition: only a local maximum of integration is an aperture. This prevents the same substrate from being counted as many overlapping observers, but Holos borrows it as a structural constraint rather than deriving it, and it inherits the unsettled question of which measure defines the maximum. Until the measure is fixed, the boundaries between observers are fixed only in principle.

Two clarifications bound this problem without solving it. First, the maximality condition individuates finite observers only; the totality never wins it, since as a whole it splits into parts that never touch. Second, the condition is silent about scale. Whether apertures can form at intermediate levels of organization, larger than any brain and smaller than everything, is left open: Holos neither asserts nor excludes them, and the same rule would govern wherever they might form.

These are not peripheral loose ends; they concern the integration threshold, one of the framework's two additions to physics. They are recorded here so that progress on them can be judged against a stated standard.

Revisions

Holos is written in public and revised when it is wrong. These claims appeared in earlier versions and have been retired or replaced, each for a stated reason.

- **Ordered dark matter.** Retired: cosmological dark matter existed before any life could, and it outweighs all the ordinary matter anything could be built from.

- **Theological and secular readings as interchangeable.** Replaced by the monist position: the structural reading is weaker, because it leaves "why am I this one?" unanswered.
- **Observation as selection.** Replaced: observation selects nothing and erases nothing; each branch is registered from within.
- **Sealing a whole branch from beginning to end.** Replaced by an observer's causal past, which gives the claim experiential content.
- **Axioms that left out the two additions.** Rebuilt: the threshold and the totality are now Axioms 3 and 5; Manifestation became the definitions of lived, lit, and unlit; Structural Constraint became a companion principle.
- **The whole causal past as "lived".** Split: lived is where experience occurs, inside observers; lit is the causal past it draws on. No one lived through the early universe.
- **Born weights as shares of experience.** Retracted: experience is not pooled. The weights are now read as self-locating odds.
- **Omega as a limit finite systems approach.** Cut: the whole exists now, the limit was never defined, and it implied a fully registered whole, against unlit structure. Omega's material moved from the extrapolations into its own core section.
- **One threshold value for every system.** Layered: the commitment is one universal shape of transition; one universal value is a stated conjecture. Neural inertia is no longer read as simple support.
- **Economy as Omega's payoff.** Replaced: one subject versus many is only a count. Omega's work is dissolving "why am I this one?" and the puzzle of which perfect copy is you.
- **Omega as the ground of record agreement.** Dropped: physics secures agreement. Omega's one job is unity: one subject, not many.
- **The bridge to integration as purely definitional.** Replaced by a small definitional core plus an argued identification that can fail.
- **Five observer requirements.** Extended to six: aboutness, tracking something beyond the system through its own channels, rules out inert high-integration arrays.
- **Current AI as "no one home."** Replaced by an open question with two borders: grounded aboutness and integration.
- **Six observer requirements.** Reduced to four: recursion and causal autonomy passed any thermostat on their own, and what they aimed at, feedback, is part of integration.
- **Aboutness grounded in senses of the system's own.** Replaced: it judged the message, not the machine, and a brain fed a perfect simulation showed it contradicting Axiom 4. Aboutness is now structural, any live channel counts, and the open question for current AI narrows to integration.
- **Anesthesia and cultured-network transitions as confirmations.** Retired as tests: ordinary models predict them too. They remain correlate probes.
- **The qubit observer-cut experiment.** Retired: qubits register nothing below the threshold, and the predicted result was ordinary contextuality.

- **"Falsified outright" for the standing bet.** Replaced by a fallback declared in advance: a consciousness-linked collapse would falsify the no-collapse physics, not the core, and the threshold would become the collapse point.
- **"Confirmed so far" for the standing bet.** Corrected to untested: no superposition tested so far has contained an observer.
- **The threshold value as a constant of nature, like the speed of light.** Replaced: a threshold that could differ between physically identical worlds would allow a copy with no one home, against Axiom 4. The conjectured value is now structural, like a percolation threshold.
- **Sharpness from a quantity "exactly zero on one side."** Replaced: that holds only in infinitely large systems. Whether anyone is home is which side of a threshold fixed by structure a system stands on; in anything finite, the outward signs blur.
- **Waking-brain criticality as evidence for the threshold's shape.** Reread: it places a critical point at the center of conscious life, where variety peaks. The threshold is where parts join into one whole; evidence for its shape must come from the boundary.
- **Aboutness as any live channel, tracked as it happens.** Sharpened: one thermometer wired into an inert array passed it, and "as it happens" excluded dreams. Now the world must be able to drive the whole across much of its range, through channels open or gated.
- **Omega exempt from maximality as "not a system among systems."** Replaced: Omega is physically the universal quantum state, so the exemption was special pleading. The whole fails integration outright, splitting into parts that never touch, which is also why it pools nothing.
- **Creation as every world the laws permit.** Narrowed: if every lawful world were real structure, reality would outrun Omega. Creation is what physics actually produces, every branch of the one quantum state; the cookbook became a buffet.
- **Omega defined through Axiom 2, and Axiom 4 at risk in the standing bet.** Corrected: Omega is the universe's complete quantum state on any reading, and Axiom 4 survives a collapse. What a collapse would cost is the claim that observation changes nothing it registers.
- **Collapse as an observer's registration.** Corrected: records are definite within each branch without observers, a detector's click included. An observer adds not definiteness but its being lived.

Holos rests on one fact and two additions. The fact: experience exists. The additions: the totality, Omega, the one experiencer; and the threshold where it wakes. Each act of experience is the totality registering itself through a local aperture. The whole is not proved from the parts; the parts are understood through the whole.

Predictions

Predictions²¹

Holos does not add new dynamical laws or modify the equations of physics. It adds two ingredients beyond them: the integration threshold Φ_c , a structural fact about where observation occurs, and the totality, Omega, as the fundamental ground of experience, of which every observer is a local aperture: an opening through which it registers itself. The commitments, expectations, and tests below follow either from established physics or from those two additions. The speculation at the end does not; it is labeled as such.

The sections below separate four kinds of claims. The last section includes speculative extensions that aim to produce observable signatures, not just philosophy.

- **Commitments:** what must be true if Holos is correct, independent of any future experiments.
- **Expectations:** patterns we should already observe in neuroscience, quantum foundations, and cosmology if those commitments are right.
- **Testability and its limits:** what cannot be tested (presence itself), a structural test that can fail, a consistency check, and a standing bet on the physics Holos adopts, with its consequences for the framework declared in advance.
- **Speculation:** extensions that *could* follow under Holos on long timescales, stated with explicit alternatives rather than predictions.

For the operational definition and the observer criteria, see Logic.

Commitments²²

Commitments 1 and 2 are fundamental to Holos. If either is rejected in principle, the framework fails as a coherent account of how reality becomes experienced. Commitment 3 rests on the no-collapse reading of Axiom 2; what losing that reading would cost is declared in advance under The Standing Bet.

1. Presence depends on observers

From Axioms 3 and 4, with the definitions in D7. See Axioms.

A physical description can be complete and still fail to explain why there is anything it is like to be inside the system it describes. The gap is not a missing fact: a perfect physical copy has the same inside. It is that the description is written from outside, and can be true and complete while saying nothing of whether anything is being lived at all.

The claim is not that observers modify physical dynamics. It is that a world becomes actualized reality only when information is registered from an internal perspective. Without registration, there is structure, but no lived fact.

| *Consistency alone does not produce presence. Presence requires registration.*

Unobserved histories therefore remain valid structures within C , the space of what physics permits, but without O , an observer to register them, they are not experienced realities.

Anthropic principles explain why observers find themselves in observer-compatible universes. They do not explain how observation itself exists or why physical structure is experienced from the inside. This framework addresses that gap.

The existence of experience demonstrates that self-registering structures are not merely abstract possibilities: physics permits them to be built, and at least one has been. Once such a structure is realizable even once, actualized reality exists, regardless of how rare or contingent its emergence may be.

Closure uses three words, each for one thing: **lived**, where experience occurs; **lit**, the causal past it draws on; and **unlit**, structure outside every observer's causal past. They are defined in [Logic](#), [D7](#).

2. Observerhood is thresholded

From Axiom 3. See [Axioms](#).

Holos rejects the idea that experience increases smoothly with greater amounts of computation. Distributed processing can scale indefinitely without producing a single point of view.

What matters is integration. Below a critical level, there is no unified internal state that could count as “what is happening for the system.” Above that level, in a system that meets the other observer requirements, experience is unavoidable.

Observerhood requires $\Phi \geq \Phi_c$

Observerhood is neither ubiquitous nor optional. It appears when structural conditions for integration are met.

Whether a system crosses this threshold is a fact about how its parts are wired together, and integration alone is not enough: the integrated state must be about a world beyond the system (see the observer requirements in [Logic](#)). It belongs to the structural layer of reality, alongside the laws of physics, and is not itself indexed to any other observer. Observerhood is what qualifies a system to have a perspective at all; it is not relative to one. In the monist reading, crossing the threshold is where an aperture opens: the totality registers itself through the system.

Two consequences follow. A physically identical copy of an observer cannot lack experience (Axiom 4). And the threshold is sharp while its surroundings are not: the fuzziness of real cases lives in richness, in the outward signs, and in our instruments, not in whether anyone is home (see [Consciousness](#)).

3. Facts are relational but consistent

From Axioms 1 and 2. See [Axioms](#).

Holos distinguishes two kinds of facts. **Structural facts** describe what is consistent: the laws of physics, the space of allowed histories with their quantum weights, and whether a system meets the integration threshold. These are absolute and observer-independent. **Registered facts** describe what is actualized as experience: which outcome a system registers from its own perspective. These are always indexed to observing systems.

The relational commitment applies to registered facts. There is no absolute, observer-independent fact about which outcome is experienced. The structural layer, by contrast, is not relative; without it, registration would have nothing stable to close against.

This does not imply contradiction, and Holos is specific about why. No possibility is erased (Axiom 2), so observers never collide over a single shared outcome. Where registrations would be incompatible, they belong to different branches of the possibility structure, each internally consistent. There is no rule that the first observer fixes the truth for everyone; relativity permits no such “first,” and none is needed.

Within a branch, consistency is operational rather than abstract: whenever two observers actually compare records, their records agree. Perspectives may differ while separated; communication forces agreement. This is the checkable content of “global consistency.” Physics secures it: observers who communicate are reached by the same signals.

Branches are weighted, not merely counted: the statistics every observer records follow the Born rule. The weights are structural facts, not amounts of experience, and for an observer they are odds of being in one branch rather than another. See [Logic](#) for the full account.

Registered facts are relative to observers. Structural facts are absolute. Observers who compare records agree.

Collapse is therefore not a new physical process. Within each branch, records are already definite, a detector's click included; what an observer adds is not the definiteness but its being lived.

Everything that follows assumes these commitments. What comes next addresses what we should expect to observe in the world if they are correct.

Expectations²³

These expectations describe what should be observed in existing domains if the commitments of Holos are correct. Most are shared with rival views, so meeting them fits Holos without confirming it; persistent failure across domains would undermine the framework.

Neuroscience: Discrete transitions in conscious access

If observerhood requires a minimum level of integration, then transitions between conscious and unconscious states should not appear as smooth signal degradation. They should resemble state changes.

Large-scale neural integration measures should therefore show the signatures of a critical transition near loss and recovery of consciousness: slowing and growing fluctuations as the boundary nears, and scaling exponents that match across individuals and species, rather than a smooth fade. These must be measured on integration at the boundary: the known near-criticality of waking cortex marks where observers operate, not the threshold. Below threshold, processing continues without unified access to experience.

If something like a Global Neuronal Workspace (a theory in which conscious access is information broadcast brain-wide) is involved, these threshold crossings should appear as sharp switches into brain-wide availability rather than a gradual fading of what can be reported.

Proxy measures such as PCI are relevant not as definitions of consciousness, but as probes of whether integration crosses a critical boundary.

A sharp transition would fit Holos but not confirm it: ordinary physicalist models predict tipping points too. What can fail is Test A; see also the [note on integration measures](#).

Quantum foundations: Observer-relative facts without collapse

If facts are brought into being through registration, quantum experiments should continue to allow descriptions in which different observers register incompatible outcomes without violating global consistency.

Holos therefore sides with branching approaches, in which no possibility is erased, and borrows one insight from relational approaches: registered facts are indexed to the systems that register them. It does not adopt Relational Quantum Mechanics itself, which rejects the universal state that branching requires. The operational signature is agreement: whenever observers within a branch compare records, the records match. A confirmed, irreconcilable record mismatch between communicating observers would falsify this commitment.

Cosmology: Ontological filtering rather than fine-tuning

The observed universe lies within the narrow range compatible with long-lived observers, not because constants were dynamically tuned, but because only such structures become experientially present.

Observer-incompatible universes, if physics actually produces them, would exist as structure while never being lived; universes the laws merely allow are possibilities, not structure. With no apertures, the totality has no opening into them, and they remain unlit structure. The nearest examples are not exotic: under the branching picture, observer-free branches of our own universe are unlit structure in exactly the same sense. Anthropic reasoning (the observation that we can only find ourselves in a universe able to support us) is therefore reframed as ontological filtering rather than selection.

This is a reframing, not a new prediction: it expects exactly what standard anthropic reasoning expects, and no cosmological observation could favor one over the other.

Minimal neural systems: emergence of coherent integration

If observerhood depends on informational integration rather than biological scale, then small biological neural networks interacting with an environment should exhibit measurable transitions in system-level coherence as integration increases.

Recent experiments with cultured neural networks connected to digital environments suggest that biological neurons can form closed feedback loops outside of a full organism. Under the Holos framework, progressively increasing connectivity, feedback richness, and environmental coupling should eventually produce a regime where neural activity shifts from distributed dynamics toward unified system-level organization.

Such transitions would not demonstrate consciousness directly. However, the existence of a reproducible boundary between loosely coupled neural computation and coherent integrated dynamics would support the claim that observerhood depends on structural integration rather than on organismal complexity. That support counts only under the conditions in the note on integration measures: a measure and threshold fixed in advance, and a stated way to lose.

Testability and Its Limits²⁴

The central claim of Holos is that observation is a closure condition, not a force: it changes no equation and moves nothing. But every experiment is a physical measurement, and an instrument only ever registers physical change. So **presence itself cannot be detected directly**. An instrument that finds nothing extra is exactly what Holos predicts, because there is nothing extra to find: presence is what the physics is like from the inside, not an additional signal beside it.

This is not a gap Holos has failed to close. It follows from the framework's own commitment that observation is dynamically inert. The sharpened form of the objection is the *unfolding argument*: for any conscious system one can in principle describe a behaviorally identical twin wired differently, and no external test could separate them. Holos accepts this. The metaphysical core (presence, and the totality it belongs to) cannot be settled by any experiment.

What remains testable is not presence but its **structural preconditions**: claims about what observation requires, and how registered facts behave. These live in the physical world and can genuinely fail. One is stated below as a test, with an explicit way for Holos to lose. The other is stated as a consistency check: its expected outcome is the one standard quantum mechanics already predicts, so it guards against contradiction rather than singling Holos out. A prediction Holos shares with rival theories cannot single it out, but a shared prediction it could fail is still worth more than one it cannot. Beneath

both sits a standing bet, stated after them, on the physics Holos adopts, with what it would cost the framework declared in advance.

Test A: Consciousness tracks integration, not behavior²⁴

Bedside assessment treats responsiveness as the sign of consciousness: a patient who follows commands is conscious, and one who does not is presumed not to be. No serious theory equates the two, but the proxy runs deep in practice. In Holos, what matters is *integration*, and integration can come apart from outward behavior. When the two diverge, Holos bets that experience follows integration.

The bet is losable because cases where the two come apart already exist. People under ketamine, in REM dreaming, or in certain seizures are behaviorally unresponsive yet later report vivid experience; sleepwalkers and some automatisms are responsive yet report little or nothing. These are natural experiments that pull integration apart from responsiveness.

Objective

Across states where responsiveness and integration diverge, determine whether later reported experience tracks a measure of integration or tracks behavioral responsiveness and arousal.

Method

Combine study designs where subjects report only afterward, or not at all, with integration proxies such as the Perturbational Complexity Index across wakefulness, anesthesia, sleep stages, and dissociative states, treating behavioral responsiveness and integration as separately varying factors rather than proxies for each other.

Held-out states, not the answer key

PCI's cutoff was set by calibrating it on people whose state was already known from their reports (Casarotto et al. 2016), and the conscious side of that calibration included waking, REM dreaming, and ketamine with vivid reports. Those cases cannot confirm Test A: they are the answer key, not the exam. The test therefore freezes the cutoff before new data and counts only states that played no part in setting it, named in advance: dream reports from non-REM sleep, sleepwalking and other automatisms, complex seizures, covert awareness in unresponsive patients detected by brain imaging, and psychedelic states. This is the design of the COGITATE collaboration: predictions fixed first, data second.

Holos Prediction

Where they diverge, reported presence follows integration: high-integration/low-responsiveness states are experienced; low-integration/ high-responsiveness states are not.

How Holos loses

The clean way to lose is a positive report from below the line: subjects who, on waking or recovering, give detailed reports of experience from periods when their integration was below threshold, by a measure and cutoff fixed in advance. A report is evidence that something was experienced, and it cannot be explained away as a failure of memory. If such reports turn up reliably, experience does not depend on integration the way Holos claims, and the framework's core structural claim fails. The reverse finding, high-integration states that yield no reports, counts against Holos only once report failure can be ruled out (see below).

The confound, and which reports count

Reports require memory, and the states this test targets are precisely those where memory is least reliable. A report of nothing is therefore ambiguous between *no experience occurred* and *experience occurred and was not encoded*. The evidence delivers unremembered; the prediction needs unexperienced. This confound is not currently controlled, so silence carries little weight in either direction: it cannot confirm absent experience in low-integration states, and it cannot refute present experience in high-integration ones.

Positive reports are unaffected, and they carry the test both ways. Where integration is high and behavior is absent, subjects report rich experience; on the held-out states above, reports of that kind would confirm the prediction without leaning on silence. Where integration is low, a positive report is the losing case described above. Holos rests Test A on positive reports, and treats silence as suggestive pending a design that calibrates report failure against states with known encoding.

The live challenge: dreaming in non-REM sleep

The losing case is not hypothetical. Awakenings from non-REM sleep often produce dream reports, yet non-REM sleep is where global measures such as PCI fall well below the waking range. If vivid reports reliably follow periods whose measured integration sat below the cutoff, Test A is lost.

Holos has one principled reply, and it comes with a condition. The maximality condition allows an aperture smaller than the whole cortex, and work on the neural correlates of dreaming finds that whether a dream is reported, in REM or non-REM sleep, tracks local activity in posterior cortex rather than the state of the brain as a whole. An aperture confined to that region could be integrated above threshold while the whole brain is not. But the reply is admissible only if the measure is local by

commitment, stated before the data are in. A measure chosen afterward to rescue the prediction would turn the test into decoration, the failure the note on integration measures below warns against.

What this can and cannot show: this tests a necessary structural condition, not presence itself. It cannot prove an integrated system is an observer, only whether integration is what experience depends on. Holos shares this prediction with other integration-based accounts of consciousness; it is a test Holos could fail, not a signature unique to Holos. It also feeds the calibration engine for the framework's open problems, with a strict division: calibration states locate the threshold and cull the candidate measures, and only held-out states test the claim (see [A path to the threshold](#)).

Check B: Observer-relative facts²⁴

This check lives in quantum foundations, not in the theory of mind, and its status comes first: it is a consistency check, not a test that could single Holos out. The outcome it anticipates is the one textbook quantum mechanics already predicts. What experiments in this family probe is the family of interpretations Holos belongs to (branching, with no absolute observed events), not Holos alone. The framework's distinctive content, which structures are present as experience, is ontological rather than experimental.

Extended [Wigner's-friend](#) experiments, in which one observer measures another observer who has already made a measurement, already pursue exactly this question. The 2020 [Local Friendliness no-go theorem](#) and its photonic tests show that if an in-lab observation counts as a genuine fact, then absoluteness of observed events, locality, and freedom of choice cannot all hold together. Holos gives up the absoluteness of observed events: registered facts are observer-relative, while structural facts and consistency remain intact.

One caveat follows from Holos's own threshold: the "friends" in current photonic tests are far below Φ_c and register nothing, so these experiments constrain the logical structure of observed events, not registration itself. Holos predicts that repeating them with genuine observers would change nothing physical, a prediction formalized as the standing bet below.

The field is moving toward genuine friends. [Wiseman, Cavalcanti, and Rieffel \(2023\)](#) proposed running a human-level artificial intelligence on a quantum computer as the friend. A September 2026 preprint, not yet peer reviewed, placed [agent-like observers on IBM quantum hardware](#) and found the violations intact. Agents that store results and predict their own measurements are still far below Φ_c by Holos's standard, so the bet remains untested, but this is the road that reaches it.

Holos Expectation

Local Friendliness inequalities keep being violated exactly as quantum mechanics predicts, at every scale the experiments reach, including as the friend is made larger and more complex.

How Holos loses

No experiment in this family can restore the absoluteness of observed events on its own: the theorem is a proof, and a violation only forces a choice among its assumptions. What could go wrong for Holos is dependence on scale. If the violations shrink or vanish as the friend grows toward a genuine observer, beyond what decoherence accounts for, observation is doing something physical, and the standing bet below is lost.

What this can and cannot show: checks Commitment 3 (facts are relational but consistent) against quantum mechanics as it is actually observed. Holos shares its expectation with standard quantum mechanics and with other views that give up absolute observed events, such as Relational Quantum Mechanics; a positive result supports the family, not Holos alone. A laboratory analog with superconducting qubits, sliced into different observer cuts, would not work as a test: by Holos's own threshold, qubit readouts register nothing, and its predicted result would be ordinary quantum contextuality, which the branching picture Holos adopts already accounts for.

The standing bet: consciousness changes nothing

The commitment that observation is dynamically inert doubles as a bet. A conscious observer and a photon produce identical physics: put an integrated system in the measuring role in place of a particle, and Holos predicts no deviation whatsoever. Superpositions lose their quantum character for thermodynamic reasons, never because someone was home. This does not make experience idle. Under Holos, experience is the inside of the physics, so when the physics does everything, experience is doing its share, not nothing.

How Holos loses: if any experiment ever finds a consciousness-linked deviation from unitary quantum mechanics (a superposition that degrades when an integrated observer registers it, beyond what ordinary decoherence accounts for), Holos's physics is falsified. Observation would be a force after all.

What that would cost, declared now. Losing the bet would falsify three things: Axiom 2 as read here (no collapse), the branching picture with its self-locating odds, and the claim that observation changes nothing it registers. It would not touch the core. The threshold (Axiom 3) would become the point where collapse happens, and so, for the first time, physically detectable. Experience would still be the inside of physical activity (Axiom 4), adding no force beyond the physics; that activity would simply include a collapse law. Omega (Axiom 5) would be the universe with its single history. Lived, lit, and

unlit would apply within that one history. Holos would adopt a collapse reading of the kind Chalmers and McQueen propose, with its threshold as the collapse point.

Losing this bet would cost Holos its physics but hand its threshold the strongest confirmation it could get, on one condition: the collapse must track integration crossing Φ_c , not size. If a superposed system above the threshold collapses while an equally large one below it does not, observers matter physically, exactly where Holos says. If collapse tracks size alone, as objective-collapse models propose, Axiom 2 falls with nothing gained: the threshold stays as it is, dynamically inert, and Omega is the universe with its single history. Either way the core stands.

This fallback is stated before any result exists, which is what separates it from a rescue. It does not make the core unfalsifiable: the core still loses through Test A, if experience turns out to track behavior rather than integration. No reading of quantum physics escapes that.

The bet has a named rival. Chalmers and McQueen (2022) propose that integrated consciousness does collapse the wave function, combining integrated information theory with a physical collapse model, and note that versions of the idea could be tested on quantum computers. Holos bets the opposite, so the same experiment would decide between them. Meanwhile quantum mechanics keeps holding as systems grow: clusters of more than 7,000 sodium atoms now show quantum interference (Pedalino et al. 2026), and the simplest version of gravity-caused collapse has been ruled out in an underground experiment (Donadi et al. 2021).

Some observer-centered frameworks quietly hope consciousness does something physical. Holos formally bets that it does not, and states in advance what losing would cost. The bet is untested so far. A century of placing ever-larger systems into superposition has found no deviation of any kind, but by Holos's own threshold none of those systems was an observer: photons, molecules, and superconducting circuits all sit far below Φ_c . That record shows quantum mechanics holding at those scales; it does not yet reach the case the bet is about. The bet is also the one standard physics makes. What makes it worth stating is that Holos, unlike views that need consciousness to act, does not hedge on it: the first experiment to put a genuine observer in the friend's role settles it, and the consequences are already on the page.

A note on the integration measures²⁴

Two further experiments might look like confirmations: a sharp integration drop under anesthesia, and cultured neural networks snapping into coherence as connectivity grows. Neither counts as a test.

A sharp transition at loss of consciousness is predicted by ordinary physicalist models too (sudden tipping points and network-wide switch-ons of the kind these systems produce anyway), so observing

one confirms nothing specific to Holos. And a cultured network almost *always* shows some nonlinear transition (neurons falling into step and cascades of activity are routine dish behavior), so an experiment that counts any such transition as success cannot fail, and an experiment that cannot fail proves nothing when it passes.

These remain useful only as **correlate probes** feeding Test A, and only under two conditions: the integration measure and the threshold value are fixed in advance, and there is a stated way to lose: the observed transition tracks a non-integration variable (arousal, metabolic rate, raw activity) rather than integration. Without a pre-committed measure and a real failure condition, a transition "somewhere" is not evidence; it is decoration.

Speculation²⁵

What follows are not predictions. They're "what if" designs that *could* emerge if the Holos framework is correct. Each is held to one standard: it must be physically possible and not extremely unlikely, and each says why it is plausible.

We know the familiar hard constraints: finite signal speed, noise, and thermodynamics. At vast scales, coherence punishes bright sprawl. Integration favors compactness, locality, and long-horizon stability.

The Holosian Scale

The Kardashev Scale ranks civilizations by energy use. The Holosian scale ranks civilizations by integration. The stages below are a map of what "advancement" looks like if coherence, not throughput, is the main objective.

H0: Fragmented

High capability, low coordination. Internal conflict, waste, and short-horizon incentives dominate. Visibility is high because broadcasting is cheap and unmanaged.

H1: Planetary Integration

The civilization becomes coherent at the scale of a world. It can coordinate, self-correct, and sustain long-term projects without collapsing into factional drift.

H2: System Coherence

Coherence survives light-lag across a star system. The civilization functions as one asynchronous system and shifts from constant broadcast to rare, directed signaling.

H3: Post-Expansion

Physical sprawl stops being the default. Exploration becomes informational first, physical only when inference fails. The outward footprint shrinks even as capability grows.

H4: Deep Integration

The civilization operates like a single high-coherence system with minimal waste and minimal leakage. External visibility collapses. What remains detectable is gravitational and thermal: the waste heat no optimization can eliminate.

H5: The Limit

Maximal coherence at minimal waste. This is a limit concept, not a stage any civilization reaches.

Why it is plausible: every stage follows from two certainties, light-speed delay and waste heat, plus one assumption: that coordination pays. The assumption is the weak link. A civilization that never learns to coordinate stays at H0, loud until it ends.

Visibility Collapse

A civilization can get more capable while becoming less visible. If its optimization target shifts from outward projection to internal coherence, it will compress, encrypt, and minimize waste. Broadcast is an early-stage habit, not a mature strategy.

On the Holosian Scale, this is the natural signature of H3–H4: rising capability with increasingly optimized and less obvious radiative signatures.

The collapse applies to light, not heat: compact systems that keep computing stay warm, for the reasons given in the [Teeming Dark](#).

Why it is plausible: efficient communication already looks like noise. A perfectly compressed signal has no repeating patterns left for an eavesdropper to spot, so the better a civilization's codes, the less its traffic stands out from the background.

Observational Regime

If a civilization is H4-level integrated, the most likely remaining footprint isn't radio or lasers. It's gravity and heat. Holos uses **Dark Node** as a label for what these systems look like from the outside: compact, ordered mass structures that minimize obvious emissions while still exporting waste heat, and staying gravitationally coupled to the universe.

In this regime, you would look for masses that are dark in visible light but carry a faint infrared excess, found through infrared surveys of individual stars and through microlensing of compact dark objects, rather than radio searches. Mass maps of whole galaxies cannot see objects this small, and no instrument can read purpose off a distant warm mass. The unavoidable search channel is warmth plus weight, not messages.

Nodes are ordinary matter that has stopped shining, not cosmological dark matter. How many there can be, and why a warm dark mass is a search channel rather than a fingerprint, is set out in the Teeming Dark.

Why it is plausible: anything that computes sheds heat, and compact systems shed it warm. Brown dwarfs and rogue planets already show that compact, dark, faintly warm masses exist and can be found.

Technology

Mesostructures

These are design sketches, not predictions: imaginative illustrations of what engineering might look like if the Integration Hypothesis holds.

The structures below are H3–H4 design patterns: compact enough to stay coherent under light-lag and thermodynamics, and consequential enough to matter without bright sprawl. They span energy generation, active coherence and computation, and long-term continuity.

Holocore

A compact, gravitationally stabilized energy mesostructure designed to supply massive, long-horizon power while exporting waste heat in thermodynamically disciplined ways.

The likeliest energy backbone for a compact civilization is its home star, harvested by nearby collectors (see [Aliens](#)). The Holocore is the step beyond: it concentrates energy density rather than surface area, converting mass into controlled output through fusion, regulated accretion, or extraction of a black hole's spin.

Purpose

- Provide sustained energy for deep-time computation and preservation
- Power large-scale modeling, shielding, and entropy management systems
- Support compact civilizational infrastructure without outward expansion
- Maintain stability across millennia with minimal maintenance overhead

Design Characteristics

- Extreme energy density per unit volume
- Controlled accretion or fusion feed systems
- Waste heat shaped, delayed, and diluted, but never eliminated. The total thermal output is set by physics, not engineering
- Compact, so it cannot hide its heat: at modest power it glows faintly in the infrared; at high power it glows brightly, and it is only as quiet as its output allows

The Holocore is infrastructure, not spectacle. If H4 integration suppresses bright sprawl, the energy backbone must be dense and long-lived, and it can be only as quiet as its power allows: compact and powerful means hot, the same physics that keeps mature systems warm.

Why it is plausible: every step is known physics; only the engineering is unknown. Fusion releases about 0.7 percent of a mass as energy. Matter falling into a rapidly spinning black hole can release roughly 30 to 40 percent, and the [Penrose process](#) can draw out a spinning black hole's rotational energy, up to 29 percent of its mass. A civilization that prizes compactness would prize the densest energy source physics allows.

Computronium Kernel

A computational core as compact as its cooling allows, built from computronium (matter arranged so that nearly every particle does useful computation) and optimized for coherent, long-horizon modeling rather than raw throughput.

This is not a data center. It is the civilization's thinking heart: where a unified world-model is maintained across centuries to millennia.

Purpose

- Maintaining a single, stable world-model across long horizons
- Long-range planning (stellar evolution, climate, existential risk)
- Decision validation and prevention of value/goal drift
- Cross-generational model consistency

Note: The Kernel may *present* as a Dark Node if coherence optimization suppresses radiative visibility. Node describes appearance, not purpose.

Why it is plausible: two known pressures meet here. Light delay rewards compactness: a signal crosses a meter in about three nanoseconds, so smaller thinks faster. Heat punishes it: power packed too densely cannot be cooled. The Kernel sits where the two balance, as today's chips already do.

Chrono Vault

A time-optimized preservation structure designed to store civilizational identity, not merely information.

Not a library or a backup, but a continuity anchor: “If we wake up in 100,000 years, how do we know who we are?”

Purpose

- Preserving value systems and canonical constraints
- Storing decision histories and their justifications
- Rebooting culture after dormancy, collapse, or fragmentation
- Anchoring identity against drift across deep time

Distinct from the Kernel: the Kernel thinks (active coherence). The Chrono Vault remembers (passive persistence).

Note: Unlike the Kernel, a Vault that stops computing can go cold. It would then not be a Dark Node, which still exports waste heat, but the sleeping case: compact, dark, and close to undetectable.

Why it is plausible: durable storage is ordinary engineering; what is speculative is the motive to pause. The [aestivation hypothesis](#) argues computing is cheaper in the colder far future, so a civilization might sleep until then; a [published reply](#) argues heat can be dumped cheaply today, so waiting is not required. Collapse, fragmentation, and dormancy remain reasons enough to keep a way back.

Communication

Under known physics, there is no scalable form of real-time interstellar dialogue. Communication converges toward transmitting large, self-contained informational payloads at light speed using extreme optical collimation.

At these distances, collaboration is necessarily asynchronous. Civilizations may contribute to shared problem spaces by exchanging durable models, partial solutions, and validated results that remain meaningful even when received centuries or millennia out of causal sync. Progress does not depend on shared present time.

Phase-Coherent Beam Transmission

Communication occurs via long-duration, phase-coherent optical channels that transmit compressed, self-describing informational payloads between known or inferred endpoints.

- **Purpose:** transfer interpretable physical, predictive, and explanatory models across interstellar or intergalactic distances, from small updates to entire civilizational knowledge bases.
- **How it works:** diffraction-limited optical beams, extreme collimation, long integration times, and heavy forward error correction referenced to invariant physical structures.
- **Payload:** layered encodings beginning with mathematics and physical constants, followed by reference frames, compression schemes, and predictive models sufficient to interpret all subsequent data.
- **Why it dominates:** photons provide maximum speed, minimal latency, and arbitrarily large total information transfer given sufficient energy and time.
- **Visibility:** unless the receiver is aligned in space, time, and frequency, the transmission is effectively invisible.

Why it is plausible: a laser sent through a ten-meter mirror spreads, across ten light-years, to a spot smaller than Earth's orbit. A beam therefore delivers far more signal per watt than broadcasting in every direction, and it is dark to everyone outside its narrow cone.

Exploration

At cosmic scales, most structure is mapped remotely and shared through long-horizon communication. Physical exploration is therefore rare, deliberate, and reserved for regimes where inference alone breaks down.

When physical probes are deployed, they are not explorers in the human sense. They are precision instruments: compact, autonomous, and built to operate alone for decades or longer.

Sentinel Probes

Highly compact, self-contained probes designed to persist in complex environments while gathering high-value physical measurements that cannot be resolved remotely.

Purpose

- Resolve observational ambiguities by direct measurement where models diverge.
- Characterize environments with nonlinear, emergent, or rapidly changing dynamics.
- Test and refine predictive models used at civilizational scale.

Technological characteristics

- Fully autonomous operation, with no expectation of real-time command or intervention.
- Onboard computation sufficient to evaluate, prioritize, and compress observations in situ.
- Preference for passive sensing and indirect interaction over active probing.
- Extreme energy efficiency enabling long dwell times with minimal thermal or electromagnetic signature.

Operational behavior

- Extended periods of quiescence punctuated by brief, targeted activity.
- No requirement for interaction with local systems or intelligences.
- Communication limited to rare, high-density transmissions rather than continuous telemetry.

Past H3, exploration scales through patience: sentinel probes exist to watch, not to arrive.

Why it is plausible: Ronald Bracewell proposed exactly this in 1960: a probe parked in a target system studies it in detail no beam can match, and can wait indefinitely. Our own spacecraft already run autonomously for decades.

Gravitational-Lens Observatories

Observation systems that exploit natural gravitational lenses to achieve extreme resolution without large, radiative infrastructure.

- **Purpose:** deep inspection of distant systems already identified as anomalous, interesting, or poorly constrained by existing models.
- **How it works:** instruments positioned along stellar or mass focal lines integrate signals over long durations, trading time for resolution.
- **Implication:** exploration shifts from surveying everything to interrogating specific questions the shared map cannot yet answer.

Why it is plausible: general relativity places the Sun's focal line beyond about 550 times the Earth-Sun distance, and NASA-funded studies have already designed a mission to image an exoplanet from there.

Citations

Citations are essential to this framework. They link to the important philosophical, scientific, and mathematical work being done by many people, both in recent years and dating back across a long history of ideas, experiments, and observations. What follows is a curated list of sources that have directly informed or inspired the views expressed here.

Overview

Introduction

[See: Overview › Introduction](#)

- 1
 - [Interpretations of quantum mechanics](#) Holos is an interpretive framework: it does not add new laws of physics, only a way to understand how a physical description turns into lived experience.
 - [Philosophy of physics](#) The study of fundamental questions about space, time, matter, and the relationship between mathematical description and what we take to be real.
 - [Structural realism](#) The view that science describes relationships between things, not what they are in themselves; Holos extends this by giving observation a role in making structure lived.
 - [Block universe](#) The view that past, present, and future exist as a four-dimensional block; Holos treats observation as what registers this structure as experience.
 - [Recursion](#) $R = C \otimes O$ is composition, not feedback: creation is what physics allows, and observation registers it from within without selecting, erasing, or changing any of it.

The Meaning of Life

[See: Overview › The Meaning of Life](#)

- 2
 - [Observer Effect](#) The disturbance of a system by the physical act of measuring it. Holos denies any further, consciousness-linked disturbance: observation changes no physics.
 - [Copenhagen Interpretation](#) Observation collapses the wavefunction into a definite state. Holos rejects collapse: evolution stays unitary, and each branch is registered from within.
 - [Quantum Darwinism](#) An environment selectively proliferates certain quantum states that become classical outcomes, observed by multiple observers.
 - [Relational Quantum Mechanics](#) The properties of quantum systems are not absolute but relative to the observer.

- Participatory Anthropic Principle The universe, as a condition of its existence, must be observed. As a "self-excited circuit", the universe requires one or more observers to bring its laws into existence.
- Von Neumann-Wigner Interpretation Consciousness causes collapse. The view Holos bets against: a consciousness-linked deviation would falsify Holos's no-collapse physics, not its core (the standing bet).

Consciousness

See: Overview › Consciousness

- 3 • Hard problem of consciousness Chalmers: why does physical activity produce felt experience at all? Holos does not answer this, but reframes it: integration is the condition under which a physical system has an inside view.
- Binding problem How distributed neural activity gives rise to unified experience; Holos frames integration as the boundary where independent parts become a single perspective.
- Neural correlates of consciousness Search for what in the brain matches experience; anesthesia switching it off, and recovery switching it back on, fit integration as the condition for being an observer.
- Global Neuronal Workspace Theory (GNWT) The theory that conscious access happens when a signal lights up and spreads across the brain; Holos treats this as a model of what gets reported, not the deeper threshold (integration) for being an observer at all.
- Integrated Information Theory Consciousness as capacity to integrate information (Φ); Holos uses integration as the threshold for observation, not a full theory of qualia.
- Qualia The subjective character of experience; Holos does not reduce qualia to Φ but treats integration as the condition for "witnessing reality from the inside."
- Panpsychism Consciousness as fundamental in matter; Holos rejects universal panpsychism in favor of a threshold ($\Phi \geq \Phi_c$) so that not everything is an observer.
- Cultured Neural Network Learning Systems Recent laboratory experiments have demonstrated that networks of cultured neurons grown on silicon substrates can be interfaced with digital environments and trained through closed-loop feedback to perform simple tasks, including interacting with video game dynamics. These systems show that biological neural tissue can form adaptive feedback loops and integrated processing structures outside a full organism. While they do not demonstrate consciousness, they provide an experimental platform for studying minimal neural integration and learning dynamics.

Spacetime

See: Overview › Spacetime

- 4 • The Big Bang The present universe emerged from an ultra-dense and high-temperature initial state.
- Accelerating Expansion of the Universe The expansion of the universe is accelerating with time.
 - Spacetime A mathematical model that fuses the three dimensions of space and the one dimension of time.
 - General Relativity Describes gravity as the warping of spacetime by mass and energy.
 - Eternalism Time as an unchanging four-dimensional block in which all moments exist tenselessly, none privileged as now.
 - Block Universe Model The view that past, present, and future exist together as one four-dimensional block. All events are fixed in spacetime; nothing, consciousness included, moves through it.
 - Relativity of Simultaneity Whether two spatially separated events occur at the same time depends on the observer.
 - The Absorber Theory Radiation is a result of both forward-in-time and backward-in-time electromagnetic waves.
 - Spacetime Interval The invariant measure of separation between two events in spacetime. For light the interval is zero, though emission and absorption remain two distinct events.
 - Null Interval A spacetime interval of zero, which occurs along light rays. A zero interval does not make two events one: emission and absorption stay distinct, and no observer can ride the light.
 - Light Cone The boundary of all possible paths that light can take from a given event, defining the causal structure of spacetime.
 - Null Geodesic The path light follows through spacetime. Its interval is zero, but it still connects two distinct events, and nothing can ride along it.
 - Retrocausality The idea that future events can influence past ones. Holos rejects it: the quantum eraser needs no backward influence, only the sorting of records made later.
 - Quantum Eraser Experiment The screen pattern never changes; interference appears only when recorded hits are sorted using later measurements. Ordinary quantum mechanics predicts every result, with nothing traveling backward in time.

Infinity

See: Overview > Infinity

- 5 • Gravitational singularity Where general relativity predicts infinite density. Most physicists read it as the place the theory stops working, the same lesson as the ultraviolet catastrophe.
- Projective Geometry A branch of geometry studying what stays the same when the point of view changes; parallel lines meet at infinity.
 - Loop Quantum Gravity A proposal that space comes in discrete quanta, which would remove the singularities of classical gravity. Unconfirmed.

- [Ultraviolet catastrophe](#) Classical physics predicted infinite radiation from hot objects. The infinity signaled a broken description, and the fix, energy in discrete packets, became the foundation of quantum theory.
- [Riemann Sphere](#) The plane plus one point at infinity, closing an unbounded surface into a finite sphere.
- [Point at Infinity](#) In projective geometry, the point where parallel lines converge, representing the boundary where infinite space folds into a finite structure.

Aliens

[See: Overview › Aliens](#)

- 6 • [Rare Earth hypothesis](#) Complex life is rare, so the silence is literal emptiness. A column in the Fermi comparison table.
- [Great Filter](#) Robin Hanson: some step stops almost every civilization before it spreads. A column in the Fermi comparison table.
- [Aestivation hypothesis](#) Civilizations sleep until the universe cools, when computing is cheaper. A column in the Fermi comparison table; contested by Bennett, Hanson, and Riedel (2019).
- [Jevons paradox](#) Efficiency gains tend to raise total consumption. The objection to the Integration Hypothesis; its answer is that light-speed delay caps one mind's useful size, so growth becomes more compact nodes near home, harvesting the home star.
- [Project Hephaistos](#) Uppsala-led search for waste heat from partial Dyson spheres around individual stars: the star-by-star channel the Integration Hypothesis points to.
- [Suazo et al. \(2024\), Project Hephaistos II: Dyson sphere candidates from Gaia DR3, 2MASS, and WISE](#) Monthly Notices of the Royal Astronomical Society: about five million stars searched; seven M-dwarf candidates with unexplained infrared excess.
- [Zackrisson et al. \(2026\), Project Hephaistos IV: JWST observations of two Dyson sphere candidates](#) Preprint, not yet peer reviewed: JWST traces the infrared excess of two candidates to background galaxies.
- [Korn et al. \(2026\), Project Hephaistos III: characterizing anomalous infrared sources](#) Preprint, not yet peer reviewed: no clear explanation yet for the remaining candidates; dust or background galaxies remain possible.
- [Gaia Data Release 4](#) ESA, due 2 December 2026: the next star-by-star census, extending the base for waste-heat searches around individual stars.
- [Fermi Paradox](#) The discrepancy between the lack of evidence for extraterrestrial life and the high likelihood of its existence. The Integration Hypothesis, a companion to Holos, reframes this silence: advancement favors compact, efficient integration over expansion and broadcast, so maturity coincides with electromagnetic quiet.
- [Early broadcasting phase](#) The early biological and broadcasting phase of a civilization. Any hurdle (abiogenesis, nuclear war, coordination failure) that stops a civilization before deep integration is

an early filter relative to true maturity.

- Light-speed latency A high-integration intelligence cannot function with years of light-speed lag between star systems. Independent interstellar colonies either fragment into less-capable outposts or the civilization turns inward, deepening local integration instead of expanding.
- Dark Nodes The hypothesized mature state: compact, non-luminous systems built from ordinary matter, a trace population detectable, if at all, by gravity and waste heat rather than light (the Teeming Dark). Not cosmological dark matter, which predates any possible life.
- Waste-heat SETI (the $\hat{G}$ survey) Wright et al. (2014): infrared searches for civilizations with large energy supplies, built on the principle that waste heat is the one emission technology cannot eliminate. This is the search channel the Teeming Dark aligns with: silent, but warm. A channel, not a fingerprint: warm dark masses are also what failed stars and cooled remnants look like.
- Ehrenfest argument Paul Ehrenfest (1917) showed that in dimensions greater than three, atomic orbitals and inverse-square planetary systems would destabilize. Matter would spiral into nuclei/stars or fly apart. Holos agrees, and goes further: higher dimensions are descriptions, not places. Nothing migrates into them (the Structural Constraint principle).
- Ephemerization R. Buckminster Fuller (1938): the process of doing “more and more with less and less” until intelligence can “do everything with nothing”. The Integration Hypothesis extends this to civilizations turning inward; Fuller did not.
- The Transcension Hypothesis John Smart (2011): advanced civilizations migrate to inner space for efficiency. Holos shares the inward-turn conclusion but not the mechanism: no transmutation or dimensional exit is proposed. Mature systems remain ordinary matter that has stopped shining.
- Cosmological natural selection Lee Smolin (1992): universes evolve to create more black holes; black hole collapse may give rise to daughter universes with slightly different constants. Cited as related work in spirit; Holos takes no position on it and proposes no mechanism linking intelligence to black holes.
- Substrate independence The view that a mind could run on different kinds of physical material, not just brains. Holos agrees that pattern matters more than material, without claiming minds can run on anything beyond ordinary matter.
- Dark matter The unexplained “missing mass” holding galaxies together. Holos takes no position on its particle nature and does not identify it with life: its fingerprints in the CMB predate stars, chemistry, and any possible builder. Mature civilizations belong instead to the non-luminous side of ordinary matter.
- Dyson sphere A hypothetical megastructure that would encompass a star to capture its energy. Their absence is consistent with the Integration Hypothesis: mature civilizations concentrate rather than sprawl. But thermodynamics still applies: waste heat, not visible structure, is the unavoidable search target.
- Brane cosmology The idea that our 3D universe may be a thin brane in a larger, higher-dimensional space. Unconfirmed; Holos takes no position, and proposes no migration into extra dimensions.

The Teeming Dark

[See: Overview › The Teeming Dark](#)

- 7 • [Black hole](#) A limited precedent for the Teeming Dark: the most compact masses need not shine and are found by gravity alone. The precedent is about compactness, not integration.
- [Massive compact halo object \(MACHO\)](#) Microlensing surveys watched millions of stars for dark compact masses and found too few to make up a large hidden population. Dark Nodes can only be a trace population.
- [Missing baryon problem](#) Surveys have now located nearly all the ordinary matter the early universe records, leaving little room for hidden built structures.

Omega

[See: Overview › Omega](#)

- 8 • [Everett \(1957\), Relative state formulation of quantum mechanics](#) Reviews of Modern Physics: the origin of the universal wave function. Physically, Holos's Omega is this one universal quantum state.
- [Everett's relative-state formulation \(Stanford Encyclopedia of Philosophy\)](#) Reference overview of Everett's theory and its interpretations.
- [Panentheism](#) The belief that the divine intersects every part of the universe and also extends beyond space and time.
- [Brahman](#) The pervasive, infinite, eternal truth, consciousness and bliss which does not change, yet is the cause of all changes.
- [Omega Point](#) A future event in which the entirety of the universe spirals toward a final point of unification. Holos borrows the name, not the idea: its Omega is the whole, not an endpoint.
- [Advaita Vedanta](#) The nondual school of Indian philosophy holding that there is one experiencer, and that each individual consciousness is that one seen through a local form; a named ancestor of the Holos monist reading.
- [Baruch Spinoza](#) Philosopher of substance monism: one substance, of which all finite things are modes or expressions.
- [George Berkeley](#) Idealist philosopher who grounded the persistence of the unobserved world in a perceiver that never looks away.
- [Open individualism](#) Daniel Kolak (I Am You, 2004): there is one person, and every one of us is it. The nearest modern relative of the Holos monist reading.
- [Vertiginous question](#) Why, of all the subjects there are, am I this one? On many separate selves it is a brute fact; on the Holos monist reading there is nothing to explain.
- [Teletransportation paradox](#) Parfit's duplication puzzle: if two perfect copies of you are made, which is you? On the Holos monist reading, both, with no remainder.

Why Are We Here?

[See: Overview › Why Are We Here?](#)

- 9 • [Unitarity](#) The principle that probabilities must sum to one, ensuring the conservation of information in quantum mechanics. Whether this holds inside black holes is the open information paradox.
- [Many-Worlds Interpretation](#) Every possible outcome of a quantum measurement occurs in a separate, branching universe.
- [Speed of Light](#) The universe's fixed speed limit. Along a light ray the spacetime interval is zero, though emission and absorption remain distinct events.
- [Indra's Net](#) An ancient Buddhist and Hindu metaphor describing an infinite web where every node is a jewel that reflects all other jewels, representing the interconnected, recursive nature of reality where each part contains and reflects the whole.

Holos

[See: Overview › Holos](#)

- 10 • [Structural Realism](#) The view that science describes the relationships between things, not what they are made of in themselves.
- [Holos](#) The whole of reality as both structure and lived experience: lawful possibility composed with registration, symbolized by ☉.
- [Recursive Operator](#) Following one observer's thread, from each registered history physics allows what comes next. Not a process in time and not feedback on physics.

Logic

Core

[See: Logic › Core](#)

- 11 • [Bekenstein Bound](#) An upper limit on the entropy or information that can be contained within a given limited region of space which has a finite amount of energy. It suggests that information is fundamentally tied to the geometry of the universe.
- [Bekenstein, J. \(2003\)](#) Information in the holographic universe. Scientific American.

Definition

[See: Logic › Definition](#)

- 12 • [Integrated Information Theory](#) IIT identifies consciousness with integrated information (Φ). Holos borrows Φ as a measure of integration only; the threshold Φ_c is its own commitment, not IIT's.

Comparison

[See: Logic > Comparison](#)

- 13 • [Goff \(2017\), Consciousness and Fundamental Reality](#) Oxford University Press: the case for priority cosmopsychism, a conscious cosmos from which individual minds derive. Holos shares the single ground but denies the cosmos any pooled experience of its own.
- [Nagasawa and Wager \(2016\), Panpsychism and priority cosmopsychism](#) Sets out priority cosmopsychism and the decomposition problem it faces: how one cosmic subject yields many individual ones. Holos's section on walled-off perspectives answers the same problem.
 - [Russellian monism](#) Physics describes structure; experience is the intrinsic nature of that structure. Holos agrees experience is the inside of physical activity, but only above the threshold.
 - [Dual-aspect monism](#) Mind and matter as two aspects of one underlying reality, from Spinoza to the present. The modern family of Axiom 4.
 - [Phenomenal concept strategy](#) The explanatory gap lies between two ways of describing one thing, not between two things. Axiom 4's floor-plan argument is a version of it.
 - [Antony \(2006\), Vagueness and the metaphysics of consciousness](#) Philosophical Studies: consciousness cannot be vague. Most who accept the argument conclude panpsychism; Holos takes the other branch, a sharp threshold that physics must locate.
 - [Rosenberg \(2004\), A Place for Consciousness](#) Oxford University Press: poses the boundary problem, what fixes where one subject ends. Holos answers with its maximality condition: one peak of integration, one perspective.
 - [Global workspace theory](#) Conscious access as brain-wide broadcast. A column in the Holos theories-of-mind table.
 - [Illusionism](#) Experience as usually conceived does not exist. A column in the Holos theories-of-mind table; Holos rejects it.
 - [Many-worlds interpretation](#) Everett (1957): every possible outcome of a quantum event really happens, in its own branch. Holos agrees all branches exist, but adds that only some are registered as anyone's actual experience.
 - [Relational quantum mechanics](#) Rovelli (1996): quantum properties are relative to observers; Holos aligns on relational facts, but keeps the universal state RQM rejects, and adds a threshold ($\Phi \geq \Phi_c$) for what counts as an observer.
 - [QBism](#) Quantum Bayesianism: quantum probabilities are agent-centered beliefs; Holos is ontological (what is lived) rather than epistemic (what agents believe).
 - [Copenhagen interpretation](#) Classical interpretation with wavefunction collapse; Holos drops collapse entirely: evolution stays unitary, branches remain, and each is registered from within.

- Objective collapse theories Theories in which collapse is a physical process; Holos rejects them. A consciousness-linked collapse would falsify Holos's no-collapse physics; its declared fallback keeps the core, with the threshold as the collapse point (the standing bet).

Primitives

See: Logic › Primitives

- 14 • Information The differentiation between possible states of a system (the difference that makes a difference).
- Phase Space The space of all possible states of a system. Creation is the space of lawful states and trajectories; Observation registers trajectories from within and selects none.
- Invariant (physics) Reality is stable relationships, not fixed properties things carry on their own.
 - ⊗ concerns where that structure is lived, not how it changes over time.

Axioms

See: Logic › Axioms

- 15 • Ontology The study of what exists. Holos separates structure, which exists whether or not it is lived, from presence, which requires observers.
- Epistemology The study of knowledge and belief. Holos distinguishes epistemic inference (what we know) from ontological presence (what is lived).

Foundations

See: Logic › Foundations

- 16 • Probability Theory ⊗ is not probability weighting: the Born weights are structural facts within C, while ⊗ concerns which structures are lived.
- Wave Function Collapse ⊗ is not a physical collapse happening over time. No outcome is picked; each branch is registered from within.
- Bayesian Inference Bayesian updating describes belief revision (epistemic). ⊗ describes how structure becomes present (ontological).
- Equivalence Relation Lit and unlit partition spacetime into two classes: a structural classification, not a transition between states.

Ontology

See: Logic › Ontology

- 17 • Universality (dynamical systems) Very different systems pass through their transitions with the same critical exponents. Holos commits to one universality class for every transition into

observerhood; one universal value of Φ_c is a separate, bolder conjecture.

- Percolation threshold A threshold fixed by structure alone: in a large square grid of randomly open bonds, a path first spans the grid at exactly half open, whatever the grid is made of. The model for the conjectured universal Φ_c : structural, true in every possible world, not a contingent constant of nature. It also models both levels of the Holos claim: each lattice has its own exact threshold, yet all share the same exponents. Finite grids round off; the threshold stays exact.
- Beggs and Plenz (2003), Neuronal avalanches in neocortical circuits Journal of Neuroscience: cascades of cortical activity follow power laws whose exponents match a known universality class. Measured at the operating point; the universality commitment needs the same kind of measurement at the threshold itself.
- Curie temperature Bulk iron gains magnetism of its own below a sharp temperature, growing smoothly from zero. The switch is exactly sharp only in the large-size limit; small samples round off. The Overview's model for the threshold: binary onset, graded richness.
- Toker et al. (2022), Consciousness is supported by near-critical slow cortical electrodynamics PNAS: waking cortex runs near the edge between stability and chaos; anesthesia and generalized seizures move it away, psychedelics move it closer. A critical point at the center of conscious life, where variety peaks, not evidence that crossing the threshold is itself a critical transition.
- Critical dynamics in spontaneous EEG predict anesthetic-induced loss of consciousness and perturbational complexity (2024) Communications Biology: criticality measures in resting EEG predict anesthetic loss of consciousness and track PCI. Like Toker et al., it measures distance from the operating point, not the threshold.
- Meisel et al. (2012), Failure of adaptive self-organized criticality during epileptic seizure attacks PLOS Computational Biology: during seizures, cortical activity departs from criticality toward hypersynchrony. Joined but not varied: the differentiation requirement fails.
- Warnaby et al. (2017), A signature of neural inertia in humans Anesthesiology: in 393 surgical patients, slow-wave activity differed between induction and emergence. The lag appeared in the EEG, not in responsiveness, so the human evidence is suggestive.
- COGITATE Consortium (2025), Adversarial testing of global neuronal workspace and integrated information theories Nature: a preregistered adversarial collaboration whose results challenged key claims of both theories. The model design for Test A.
- What makes a theory of consciousness unscientific? (2025) Nature Neuroscience: argues that IIT's core identity claim is untestable. Holos does not adopt that claim and borrows Φ only as a measure.
- Aaronson (2014), Why I Am Not An Integrated Information Theorist Shows that very simple, inert structures can score higher on integration measures than a brain. Holos answers with the aboutness requirement: an observer's variety must be variety the world can drive through its channels. Wiring one sensor into such an array leaves the rest of its repertoire sealed.
- Harnad (1990), The symbol grounding problem Physica D: symbols connected only to other symbols are not about anything until some of them ground out in perception. Holos reads it

structurally: a system is about something when its states track it through a live channel of its own, whatever the channel carries.

- Friedman et al. (2010), Evidence for neural inertia PLoS ONE: in animals, consciousness is lost and regained at different anesthetic levels. Such a lag marks an abrupt, first-order switch: it supports a sharp threshold but counts against a universal shape of transition, unless it is pharmacological. Human evidence is still suggestive.
- Ontology The study of what exists. Holos separates structure, real whether or not it is lived, from experience, which occurs only where $\Phi \geq \Phi_c$.
- Causality How events bring about other events. In Holos, observation causes nothing extra: experience is the inside of physical causation, not an added cause.
- Quantum Decoherence The process by which quantum systems interact with their environment. Decoherence yields classical-looking branches; Φ marks where within them experience occurs.

Relationship to Physics

See: [Logic > Relationship to Physics](#)

- 18 • Unitarity (physics) Quantum mechanics requires unitarity. Holos preserves it fully: nothing collapses, nothing is selected, and unobserved branches remain in Creation as unlit structure.
- Hilbert Space The mathematical space of all possible quantum states. Holos deletes no branches from it; the Born weights are structural facts about it.
- Schrödinger Equation Φ does not replace or modify the Schrödinger equation. It marks where experience occurs and leaves the quantum math untouched.
- Quantum Mechanics Holos leaves quantum mechanics untouched. Φ marks where registration occurs; it adds no constraint to the physics.
- Born rule Max Born (1926): quantum probabilities are squared amplitudes, the most precisely confirmed rule in physics. In Holos the weights are structural facts within C, fixing the statistics each observer records rather than measuring how much experience a branch carries.
- Gleason's theorem Andrew Gleason (1957): any consistent assignment of probabilities to quantum outcomes must take the Born-rule form: exactly one weighting is possible. Its assumptions remain debated; Holos adds no new mathematics.
- Sebens and Carroll (2018), Self-locating uncertainty and the origin of probability in Everettian quantum mechanics British Journal for the Philosophy of Science: after branching and before looking, an observer is uncertain which branch they are in, and the Born weights are the rational odds. Holos adopts this reading of the weights and reaches it by elimination: counting fails, weight survives.
- Ontology Experience is lived only inside observers; the causal pasts they draw on are lit; the rest remains unlit structure. The underlying quantum math is unaltered.

Notation

See: [Logic](#) › [Notation](#)

- 19
- Information Theory Sending information assumes a causal signal passes between places; $\otimes$ sends nothing. It concerns which histories are lived, not which signals travel.
 - Measurement in Quantum Mechanics Measurement models the physical interaction between systems; Observation in Holos is registration from within. It selects nothing and alters no interaction.
 - Hilbert Space In modern physics, the "state" of any complex system is defined as a vector in a high-dimensional space. Our perception of 3D space is a specific observable projection of this deeper geometric reality.

Companion Principle

See: [Logic](#) › [Companion Principle](#)

- 20
- Ephemerization R. Buckminster Fuller (1938): the process of doing more with less until intelligence can do everything with nothing. The Integration Hypothesis extends this to civilizations turning inward; Fuller did not.
 - Ehrenfest argument Paul Ehrenfest (1917) showed that in dimensions greater than three, atomic orbitals and inverse-square planetary systems would destabilize. Holos agrees, and goes further: higher dimensions are descriptions, not places. Nothing migrates into them (the Structural Constraint principle).

Predictions

Introduction

See: [Predictions](#) › [Introduction](#)

- 21
- Dynamics (physics) Holos does not propose new dynamical laws; it makes structural claims about how reality becomes lived ($R = C \otimes O$).
 - Ontology Predictions about how structure becomes present as experience. Observers register the block universe from within; they do not hold it together.
 - Block universe The structural layer of Holos: tenseless and observer-independent. Observers register it from within; its consistency does not depend on them.
 - Anthropic principle Holos reframes anthropic selection as ontological filtering: observer-free universes may exist as structure but are never lived.

- Cosmic microwave background (CMB) polarization CMB-S4, LiteBIRD: searching the polarization of the earliest light for primordial gravitational waves, a test of inflation. Holos makes no prediction here.
- Past hypothesis The universe began in a highly ordered, low-entropy state. In Holos, branches where nothing could live remain unlit: real as structure, never lived (no system reaches Φ_c).
- Inflation (cosmology) A proposed early phase of rapid expansion that would explain the universe's flatness and smoothness. Holos takes no position on it.
- Multiverse Branches where nothing could live are real as structure, but in Holos they are never lived (no system reaches Φ_c).

Commitments

See: Predictions › Commitments

- 22
- Integrated Information Theory (IIT) Holos borrows Φ as a measure of integration, not IIT's identity claim. Crossing Φ_c is where experience occurs; PCI is a practical proxy.
 - Panpsychism Holos distinguishes from universal panpsychism (everything conscious) and illusionism (consciousness is illusion).
 - Illusionism (philosophy) The view that consciousness is an illusion. Holos rejects it: experience is the one fact the framework starts from.
 - Qualia The felt character of experience. Holos locates qualia in systems that meet the observer requirements; below Φ_c there is processing without experience.
 - Perturbational Complexity Index (PCI) IIT-inspired metric with an empirically calibrated cutoff between conscious and unconscious states. Holos treats it as a proxy for integration, not a detector of presence.
 - Phase transition Holos treats the onset of experience as a sharp transition with one universal shape, not a dial. A sharp change alone would not confirm this, since ordinary models predict tipping points too.

Expectations

See: Predictions › Expectations

- 23
- Extended Wigner's Friend experiments Two observers can hold different registered facts about the same event without breaking unitarity. In Holos each fact is indexed to its branch and observer.
 - Unitarity (physics) Conservation of all possibilities; Holos predicts agreement among communicating observers without objective collapse.
 - Relational quantum mechanics Holos borrows RQM's point that facts are indexed to observing systems, but sides with branching, which keeps the universal state RQM rejects.

- [Objective collapse theories](#) Holos rejects objective collapse: evolution is unitary, and apparent collapse is registration within a branch.

Testability and Its Limits

[See: Predictions › Testability and Its Limits](#)

- 24 • [Chalmers and McQueen \(2022\), Consciousness and the collapse of the wave function](#) The named rival to the standing bet: integrated consciousness collapses the wave function, testable in principle on quantum computers. Holos bets it does not.
- [Wiseman, Cavalcanti, and Rieffel \(2023\), A thoughtful Local Friendliness no-go theorem](#) Quantum: proposes a human-level AI on a quantum computer as the friend, the road toward a genuine observer in Check B.
- [Laux and Cavalcanti \(2026\), Extended Wigner's friend scenarios with agent-like observers on quantum computers](#) Preprint, not yet peer reviewed: agent-like observers on IBM hardware; Local Friendliness violations persist. The agents are far below Φ_c .
- [Pedalino et al. \(2026\), Probing quantum mechanics with nanoparticle matter-wave interferometry](#) Nature: quantum interference of sodium clusters of more than 7,000 atoms, the current size record.
- [Donadi et al. \(2021\), Underground test of gravity-related wave function collapse](#) Nature Physics: rules out the parameter-free Diósi-Penrose collapse model.
- [Wigner's friend](#) Registered facts are observer-indexed; no objective collapse. Extended Wigner's-friend experiments constrain the family of views Holos belongs to (Check B).
- [Bong et al. \(2020\), A strong no-go theorem on the Wigner's friend paradox](#) Nature Physics: the Local Friendliness theorem and its photonic test. Absoluteness of observed events, locality, and freedom of choice cannot all hold; Holos gives up the first.
- [Siclari et al. \(2017\), The neural correlates of dreaming](#) Nature Neuroscience: dream reports occur after awakenings from both REM and NREM sleep, tracked by local activity in posterior cortex. The live challenge for Test A.
- [Phase transition](#) If the threshold is a genuine critical point, it should leave measurable signatures near the boundary; see A path to the threshold.
- [Casarotto et al. \(2016\), Stratification of unresponsive patients by an independently validated index of brain complexity](#) Annals of Neurology: sets PCI's cutoff on a benchmark of states known from report, including REM dreaming and ketamine. Those cases are therefore calibration, not confirmation; Test A counts only held-out states named in advance.
- [TMS-EEG](#) PCI is computed from TMS-EEG responses; it is the integration proxy Test A relies on.
- [Perturbational Complexity Index \(PCI\)](#) Validated across sleep, anesthesia, and disorders of consciousness; Test A uses it to separate integration from responsiveness.
- [Propofol / BIS index](#) Anesthesia depth: one of the states Test A compares. A drop in integration alone confirms nothing specific to Holos.

- Recurrent neural network Recurrent architectures; relevant to whether artificial systems have the feedback that integration requires.
- Neuromorphic engineering Brain-like hardware with feedback; a candidate substrate for meeting the observer requirements.
- Causal density Integration proxy: how much a network's parts predict one another's activity over time, used when computing Φ directly is infeasible.
- Collective intelligence Social and agent networks. Whether group-scale apertures can form is left open: Holos neither asserts nor excludes them.
- Mutual information Integration proxy: mutual information across subgroups, causal density, network-wide coherence.
- Network theory Small-world and scale-free structure; candidate settings for measuring integration.
- Relational quantum mechanics Holos shares RQM's observer-indexed facts but not its rejection of a universal state; a positive Check B result supports the family, not Holos alone.

Speculation

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- Phase transition A speculative reading: civilizational integration may advance in jumps rather than smoothly, as light-speed delay and waste heat reshape what can be coordinated.
 - Ephemeralization Doing more with less. The Integration Hypothesis extends this to advancement as inward growth rather than outward expansion; Fuller did not.
 - Fermi paradox A companion hypothesis to Holos, not part of its core: the Integration Hypothesis and Visibility Collapse. Mature civilizations are silent in light; the unavoidable observables are gravity and waste heat.
 - Hanson et al. (2021), If loud aliens explain human earliness, quiet aliens are also rare The Astrophysical Journal: the grabby aliens model. Visible, expanding civilizations exist but have not reached us, and we are early. Named on the site as the main rival to the Integration Hypothesis.
 - Griffith et al. (2015), The $\hat{G}$ infrared search, III The Astrophysical Journal Supplement: of about 100,000 galaxies surveyed with WISE, none hosts a civilization reprocessing more than 85% of its starlight into waste heat.
 - Zoo hypothesis The idea that mature civilizations deliberately leave young ones alone. For Holos, non-contact needs no agreement: leaving a young civilization alone costs nothing, while contact takes effort.
 - Weakly interacting massive particles (WIMPs) A leading dark-matter particle candidate. Holos takes no position on the particle nature of cosmological dark matter, which predates life; Dark Nodes are non-luminous ordinary matter.
 - Euclid Mission March 2025 Q1 data: 26M galaxies, precision mass mapping. Compact mass peaks with weak visible counterparts become interesting only after conventional explanations fail, and

only alongside an infrared excess.

- [James Webb Space Telescope \(JWST\)](#) Deep-field mass structure under active study. For Holos, a candidate Dark Node requires gravity plus faint warmth, not gravity alone.
- [Baryon](#) Ordinary matter. Dark Nodes remain baryonic: built matter that stops shining, not matter that changes kind. No transmutation is proposed.
- [Navarro–Frenk–White profile](#) Halo-profile deviations have viable conventional explanations (baryonic feedback, mergers, measurement limits). Holos claims no dark-matter anomaly.
- [Lambda-CDM model](#) The standard cosmological model. Holos does not modify it: cosmological dark matter is primordial and predates any possible life.
- [Dark Energy Survey \(DES\)](#) Jan 2026 final analysis: an open clustering tension. Holos offers no interpretation of it.
- [JWST COSMOS-Web](#) Granular mass structure under study. A Dark Node candidate would need compact mass plus infrared warmth; mass alone is not evidence.

Technology

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- [Penrose process](#) Extracting a spinning black hole's rotational energy, up to 29 percent of its mass: the known physics behind the Holocore's densest option.
 - [Sandberg, Armstrong, and Ćirković \(2017\), The aestivation hypothesis](#) Argues advanced civilizations might sleep until the universe cools, when computing is cheaper. One motive for the Chrono Vault's sleeping case.
 - [Bennett, Hanson, and Riedel \(2019\), Comment on the aestivation hypothesis](#) Foundations of Physics: argues entropy can be disposed of cheaply today, so waiting is not required. The counterweight to aestivation.
 - [Bracewell \(1960\), Communications from superior galactic communities](#) Nature: proposes parking autonomous probes in target star systems. The ancestor of Sentinel Probes.
 - [Turyshv et al. \(2020\), Direct multipixel imaging and spectroscopy of an exoplanet with a solar gravity lens mission](#) A NASA-funded mission design using the Sun's gravitational focus, beyond about 550 times the Earth-Sun distance. The basis for Gravitational-Lens Observatories.
 - [Computronium](#) Hypothetical material optimized for computation; the Computronium Kernel is a maximally compact core for coherent, long-horizon world-modeling.
 - [Megastructure](#) Large-scale artificial structures; Holos mesostructures (Kernel, Chrono Vault) are compact and coherence-optimized rather than maximally expansive.
 - [Gravitational lens](#) Bending of light by mass; gravitational-lens observatories use natural lenses for extreme resolution without large radiative infrastructure.
 - [Time capsule](#) The Chrono Vault extends this idea to civilizational identity: preserving value systems, decision histories, and continuity across deep time.

- Interstellar communication At cosmic scales, communication converges on phase-coherent optical payloads and compressed, self-describing models rather than real-time dialogue.
- Space probe Sentinel Probes are compact, autonomous, long-duration instruments that resolve observational ambiguities and operate without real-time control.